Early Warning and Prediction of Respiratory Virus Outbreaks Using Public Health Surveillance Data

Master's in Data Analytics

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Declaration

I, Rahul Patil (20057979), a student at Dublin Business School pursuing a Master's in Data Analytics, hereby declare that the work presented in this thesis titled "Early Warning and Prediction of Respiratory Virus Outbreaks Using Public Health Surveillance Data" is my original work and has not been submitted for any other degree at this or any other institution.

This thesis represents my independent effort and intellectual contribution to the field of public health surveillance and outbreak prediction. The information derived from the literature has been duly acknowledged in the text, and a comprehensive list of references is provided. The thesis adheres to all academic standards and guidelines set forth by Dublin Business School.

Signed: Rahul Patil

Date: 11 May 2026

Acknowledgment

I am immensely grateful to my project guide, Dr Shubham Sharma of Dublin Business School, for her unwavering support and invaluable guidance throughout this research journey. Her insightful perspective, constant encouragement, and motivating attitude have been a continuous source of inspiration, shaping not only this research but also my academic and professional outlook. It has been a great privilege and honour to work under her guidance.

I would also like to sincerely thank her for her kindness, empathy, and great sense of humour, which made this journey more encouraging and meaningful.

In conclusion, I express my heartfelt appreciation to Dr Shubham Sharma for her instrumental role in shaping and enriching this research endeavour. Her mentorship and guidance have been pivotal to my academic and professional development, and I am truly grateful for the opportunity to learn under such an esteemed educator.

Abstract

This dissertation examines how public health surveillance data can provide early warning of rising Influenza activity and predict the risk of an outbreak in the subsequent week. Using data from primary care sentinel surveillance, the study focuses on key indicators, including tests, detections, and positivity rates. After cleaning and preparing the data, trends and patterns that might signal an outbreak were explored. Time-based, lag-based, and rolling features were engineered to enhance predictive performance. Five machine learning models were trained and compared, with validation steps including time-series cross-validation and hyperparameter tuning to ensure robust results. All models performed well, but Random Forest emerged as the best overall. The most important predictors were positivity, recent positivity trends, and seasonal timing. These findings suggest that routine surveillance data can be leveraged in practical, interpretable ways to provide short-term warnings of influenza outbreaks.

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CHAPTER 1: INTRODUCTION

1.1 Background and Context of the Study

Respiratory virus outbreaks remain an important public health concern because they can rapidly increase illness levels, strain healthcare services, and pose challenges for surveillance and response systems. Among these viruses, Influenza remains especially important because of its seasonal recurrence, its widespread impact on populations, and the need for timely monitoring of rising activity. Public health surveillance systems are therefore used to track changes in testing, detections, and related indicators over time. Earlier work has shown that surveillance systems remain central to Influenza monitoring, while newer and nontraditional data sources are generally seen as useful complements rather than replacements for established public health reporting systems (Hammond et al., 2022).

In practice, surveillance data is often used mainly to describe what is currently happening or what has already happened. However, public health decision-making increasingly benefits from earlier warning of rising virus activity. Research has shown that some surveillance indicators can signal increasing Influenza activity earlier than broader syndromic measures. For example, Baumbach et al. (2009) found that the proportion of positive rapid Influenza test results began rising earlier than the percentages of influenza-like illness visits across multiple seasons, suggesting that positivity-related indicators may offer useful early signals of outbreak development. This makes positivity an especially important variable when studying short-term outbreak warning.

At the same time, surveillance indicators must be interpreted carefully. Recent work has shown that commonly used Influenza surveillance measures, including test-positive proportion and other routine indicators, can be influenced by factors such as testing behaviour, healthcare-seeking patterns, age differences, and the circulation of other pathogens with similar symptoms (Eales et al., 2024**.** In addition, comparative work by Mellor et al. (2024) showed that different surveillance and administrative data sources can yield distinct pictures of epidemic timing and scale, especially when examined across geographies. These studies suggest that surveillance data is highly valuable, but it also needs careful analytical treatment if it is to be used for early warning and predictive modelling.

In recent years, machine learning and forecasting approaches have been increasingly explored as ways to improve surveillance-based outbreak intelligence. Studies have shown that combining historical epidemiological patterns with machine learning can improve real-time or short-term Influenza monitoring and forecasting (Santillana et al., 2016; Cheng et al., 2020). More recent work has also shown that early-warning systems based on surveillance data can perform strongly when designed specifically for outbreak detection rather than simple retrospective analysis (Yang et al., 2024). These developments suggest that official surveillance indicators may be used not only for descriptive reporting but also for predictive public health support.

1.2 Identification of the Problem

One of the main challenges in respiratory virus surveillance is that outbreaks are often recognised only after they are already evident in routine reports. Surveillance systems generate valuable information, but the data is not always transformed into actionable early-warning insight. As a result, public health response may depend heavily on retrospective interpretation rather than proactive detection.

This is especially important in the case of respiratory viruses, as their activity can increase rapidly over a short period. A gradual change in positivity rate or the number of positive tests may appear small at first, but over a few reporting periods, it may develop into a clear outbreak pattern. If such changes are not identified early, public health agencies may lose valuable time for preparedness, communication, and planning.

Another practical issue is that routine surveillance indicators can be difficult to interpret when viewed separately. For example, the number of positive cases alone may rise because more tests were conducted, while the positivity rate may offer a different view of disease activity. Similarly, one week of increase may not always signal a real outbreak, but repeated increases across consecutive periods may be more meaningful. This suggests that outbreak warning should not rely on a single variable in isolation. Instead, it should consider multiple surveillance indicators together and examine how they behave over time.

1.3 Research Gap

One clear gap is that many existing studies treat surveillance and prediction as separate tasks. Descriptive studies often explain what happened but stop before developing a predictive early-warning framework. On the other hand, predictive studies sometimes focus primarily on model performance, without paying sufficient attention to how outbreak-related indicators behave in the data before prediction.

A second gap relates to interpretability. In public health practice, it is not enough to say that a model predicts higher outbreak risk. Decision-makers also need to understand which surveillance indicators are driving the alert. If a model produces strong performance but offers very little practical explanation, its use in routine monitoring may be limited.

A third gap is that relatively fewer studies combine the following elements within one framework: exploratory data analysis of surveillance trends, feature engineering from routine surveillance indicators, early warning signal detection, comparison of multiple machine learning models, and interpretation of the most important predictive factors. This study is designed to address that combined gap.

1.4 Aim of the Study

The aim of this study is to develop an interpretable machine learning-based framework for the early warning and prediction of respiratory virus outbreaks using public health surveillance data.

1.5 Research Questions

This study is guided by the following research questions:

1. What temporal and surveillance patterns are associated with rising Influenza activity in public health surveillance data?
2. Which surveillance indicators are most useful for identifying early outbreak signals of Influenza?
3. Which machine learning model performs best for predicting next week's influenza outbreak risk from surveillance data?
4. How interpretable and practically useful are the model outputs for supporting early warning and public health decision-making?

1.6 Objectives of the Study

The objectives of this study are to:

Objective 1) Collect and prepare official Influenza surveillance data from a public health source.

Objective 2) Analyse the dataset to identify key temporal patterns, surveillance trends, and possible outbreak signals.

Objective 3) Construct predictive features from surveillance indicators, including positivity-based, time-based, lag-based, and rolling measures.

Objective 4) Define an appropriate next-week outbreak target for classification analysis.

Objective 5) Apply, tune, and compare five machine learning models for Influenza outbreak prediction.

Objective 6) Interpret the most important predictive variables and assess their practical relevance for early warning and public health decision-making.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter reviews previous studies related to respiratory virus surveillance, outbreak detection, surveillance indicators, and machine learning-based prediction. The review is organised thematically, moving from general surveillance systems to outbreak indicators, early-warning studies, machine learning applications, and model evaluation. At the same time, each paper is discussed individually, with its method, modelling approach, challenges, and findings clearly linked to the present dissertation.

2.2 Respiratory Virus Outbreaks and Public Health Surveillance

Hammond et al. (2022) reviewed traditional and non-traditional influenza surveillance systems using databases such as PubMed, Scopus, and Web of Science. The study did not use machine learning because it was a scoping review. A key challenge was that non-traditional sources were not equally reliable across all settings. The results showed that electronic health records and participatory surveillance were useful as complementary sources. However, the authors concluded that these systems should support traditional surveillance rather than replace it.

Mellor et al. (2024) compared four influenza-related data sources in England to estimate epidemic size, timing, and growth rate. This was not a machine learning study, but a comparative surveillance modelling study. The main challenge was that different data sources gave different pictures of the same influenza season, especially at the regional level. The results showed that the choice of surveillance source strongly affects how an epidemic is interpreted. This paper is important because it shows that dataset selection matters before modelling even begins.

Bardsley et al. (2023) used a retrospective observational approach, drawing on laboratory, clinical, hospital, and syndromic RSV surveillance data. The study applied time-series analysis and generalised linear models, but not machine learning. A major challenge was the disruption caused by COVID-19 restrictions, which changed normal RSV seasonality. The results showed that RSV almost disappeared in one season and then returned with an unusual summer surge.

2.3 Surveillance Indicators, Positivity Rate, and Time-Based Trends

Baumbach et al. (2009) compared rapid influenza test positivity with influenza-like illness visit percentages across three seasons. The study did not use machine learning, as it focused on comparing surveillance indicators. The main challenge was whether traditional syndrome-based surveillance was timely enough for early warning. The results showed that positive rapid influenza tests rose earlier than clinical ILI percentages in all seasons. This suggests that positivity-related indicators can be useful for early detection of outbreaks.

Kim et al. (2022) analysed Korean national sentinel surveillance data to compare RSV and influenza activity after the COVID-19 intervention period. This was a descriptive surveillance study, not a machine learning study. A major challenge was that the pandemic had changed the normal seasonal behaviour of respiratory viruses. The study found that influenza activity remained very low, while RSV positivity increased strongly. This showed that different viruses can behave very differently under the same surveillance conditions.

Eales et al. (2024) examined common influenza surveillance indicators, including the influenza-like illness rate, the laboratory-confirmed case rate, and the test-positive proportion. The study used a theoretical and analytical approach, not machine learning. The main challenge discussed was that these indicators can be biased by testing behaviour, age differences, and other circulating pathogens. The study concluded that no surveillance indicator is a perfect measure of true infection incidence.

2.4 Early Warning and Outbreak Detection Studies

Spaeder and Fackler (2012) used retrospective time-series forecasting to predict RSV incidence at community, hospital, and intensive care unit levels. The study did not use modern machine learning models; instead, it used statistical forecasting methods. One challenge was forecasting across settings with very different case volumes. The results showed that 2-week forecasts were reasonably accurate across the three levels of care.

Reis et al. (2019) developed a forecasting model for RSV outbreaks in the United States at the national, regional, and state levels. The method combined three forecasting models into a single ensemble, akin to a machine learning ensemble. The main challenge was that the timing and intensity of the outbreak differed greatly across locations. The results showed that the super ensemble improved forecast performance, especially at the regional and state level. This paper shows that combining models can improve outbreak forecasting.

Yang et al. (2024) developed a deep learning-based early-warning model, SEAR, for influenza surveillance in China. The study compared the model with Logistic Regression, Support Vector Machine, Random Forest, and XGBoost. A key challenge was improving early-warning performance across regions while maintaining system stability. The results showed that the deep learning model outperformed the comparison models in the reported setting.

2.5 Machine Learning for Influenza and Respiratory Surveillance Prediction

Santillana et al. (2016) used electronic health records together with historical epidemiological data to estimate regional influenza activity in real time. The paper introduced the ARES model, which was based on a Support Vector Machine framework. The main challenge was the delay in traditional influenza surveillance reporting. The results showed that the model produced accurate near-real-time influenza estimates across regions. This study shows that machine learning can improve the timeliness of surveillance estimates.

Lu et al. (2018) used internet-based data sources and ensemble-style methods to monitor and forecast influenza in Boston. Their method combined multiple sources and applied models, such as ARGO, for local nowcasting and forecasting. A major challenge was generating accurate influenza estimates at a fine local scale where routine data may be limited. The results showed very high correlations for both current-week and 1-week-ahead forecasting. This paper supports the idea that combining multiple sources can improve flu prediction.

Cheng et al. (2020) built a forecasting framework in Taiwan using ARIMA, Random Forest, Support Vector Regression, and Extreme Gradient Boosting, followed by a stacking ensemble. The challenge was that influenza patterns in Taiwan are more complex because of the subtropical setting. The study found that all models performed well, but the ensemble model gave the best forecasting accuracy. This suggests that combining multiple algorithms can improve reliability.

Yang et al. (2023) combined virological surveillance data with Baidu search query data to predict influenza trends in China. The method used Pearson correlation, a distributed lag nonlinear model, and a GRU deep learning model with attention mechanisms. A challenge in the study was that search behaviour was not equally useful in all regions. The results showed that selected search terms could help predict influenza-positive rates 2 to 3 weeks in advance in southern China. This paper shows the value of hybrid surveillance, but also its regional limitations.

2.6 Model Evaluation, Forecasting Value, and the Suitability of Surveillance Systems

Reich et al. (2019) evaluated a real-time multi-model influenza forecasting ensemble in the United States. Their method combined 21 component models using a stacking-based weighted ensemble. A major challenge was that influenza forecasting performance changes across seasons and targets. The study found that the ensemble outperformed any single model during the 2017–2018 season. This paper is important because it shows that model comparison and ensemble methods are central to strong forecasting performance.

Maroufi et al. (2025) proposed a framework for assessing which surveillance systems are suitable for AI and machine-learning-based short-term influenza forecasting. The study used a two-phase design comprising document review and a structured system evaluation. It did not train machine learning models directly, but focused on surveillance suitability for future forecasting. A key challenge was that surveillance systems differ in timeliness, completeness, and representativeness. The paper concluded that not every surveillance system is equally suitable for prediction, which strongly supports the importance of choosing the right dataset.

CHAPTER 3: RESEARCH METHODOLOGY

This chapter explains the methodology used in the study. It describes the research design, data source, preprocessing steps, feature engineering process, target variable, machine learning models, and the evaluation strategy for next week's influenza outbreak prediction. The machine learning models are then introduced, followed by the validation strategy, hyperparameter tuning approach, and the final test evaluation procedure.

3.1 Data Source and Dataset Description

The primary dataset used in the study was the sentinelTestsDetectionsPositivity.csv file. This dataset contains aggregated weekly respiratory virus surveillance data reported through a primary care sentinel surveillance system. It was chosen because it is well-suited to the aim of early outbreak warning in the community and provides wider country coverage than the alternative severe acute respiratory infection dataset.

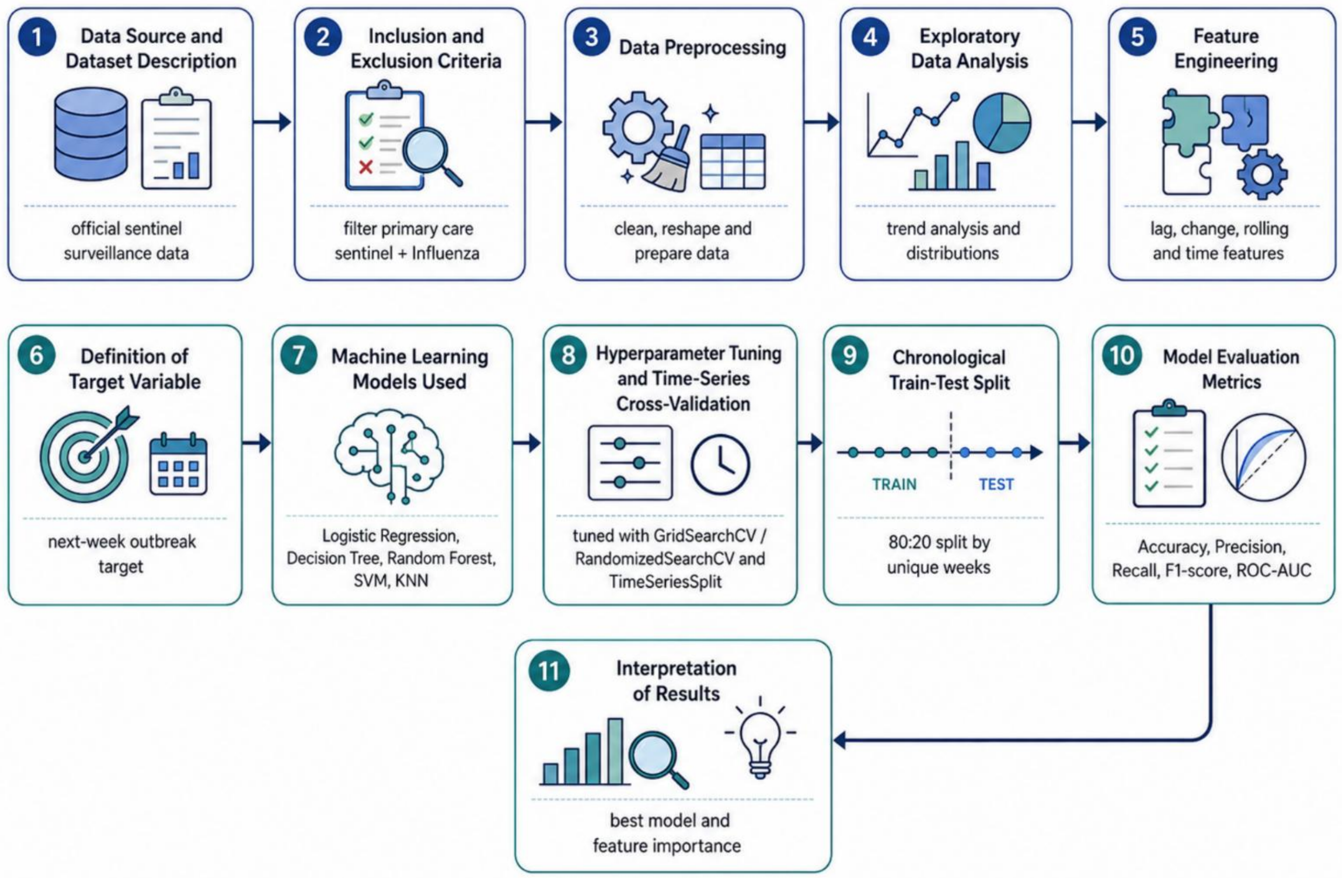


Figure 3.1.1: Overall Research Methodology Framework

The dataset includes multiple respiratory pathogens, including Influenza, RSV, and SARS-CoV-2. However, after reviewing the data structure and modelling performance across pathogens, Influenza was selected as the main modelling focus of the dissertation. This decision was made because Influenza provided the strongest and most reliable outbreak-related patterns within the available surveillance data and produced the clearest predictive results within the scope of the study.

The main columns available in the dataset are:

- survtype
- countryname
- yearweek
- pathogen
- pathogentype
- pathogensubtype
- indicator
- age
- value

The most important variables for the final analysis were the indicator-based values representing:

- tests
- detections
- positivity

These indicators were later reshaped into separate columns during preprocessing. The dataset therefore provided the key information needed for the exploratory analysis and for building a machine learning framework for next-week outbreak prediction.

3.2 Inclusion and Exclusion Criteria

The inclusion criteria were as follows:

- Only records from the primary care sentinel surveillance system were retained.
- Only records for Influenza were selected.
- Only records with the required weekly reporting information were kept.
- Only records relevant to the main indicator variables used in the study were retained.

The exclusion criteria were as follows:

- Records from other pathogens were removed from the main analysis.
- Records with incomplete key indicator information after reshaping were excluded if they could not support further analysis.
- Rows with missing lag-based modelling features were removed before final model training.
- Rows that could not support the creation of next week's target were excluded from the final supervised modelling stage.

These criteria were important because they helped keep the modelling dataset focused, consistent, and aligned with the aim of the dissertation.

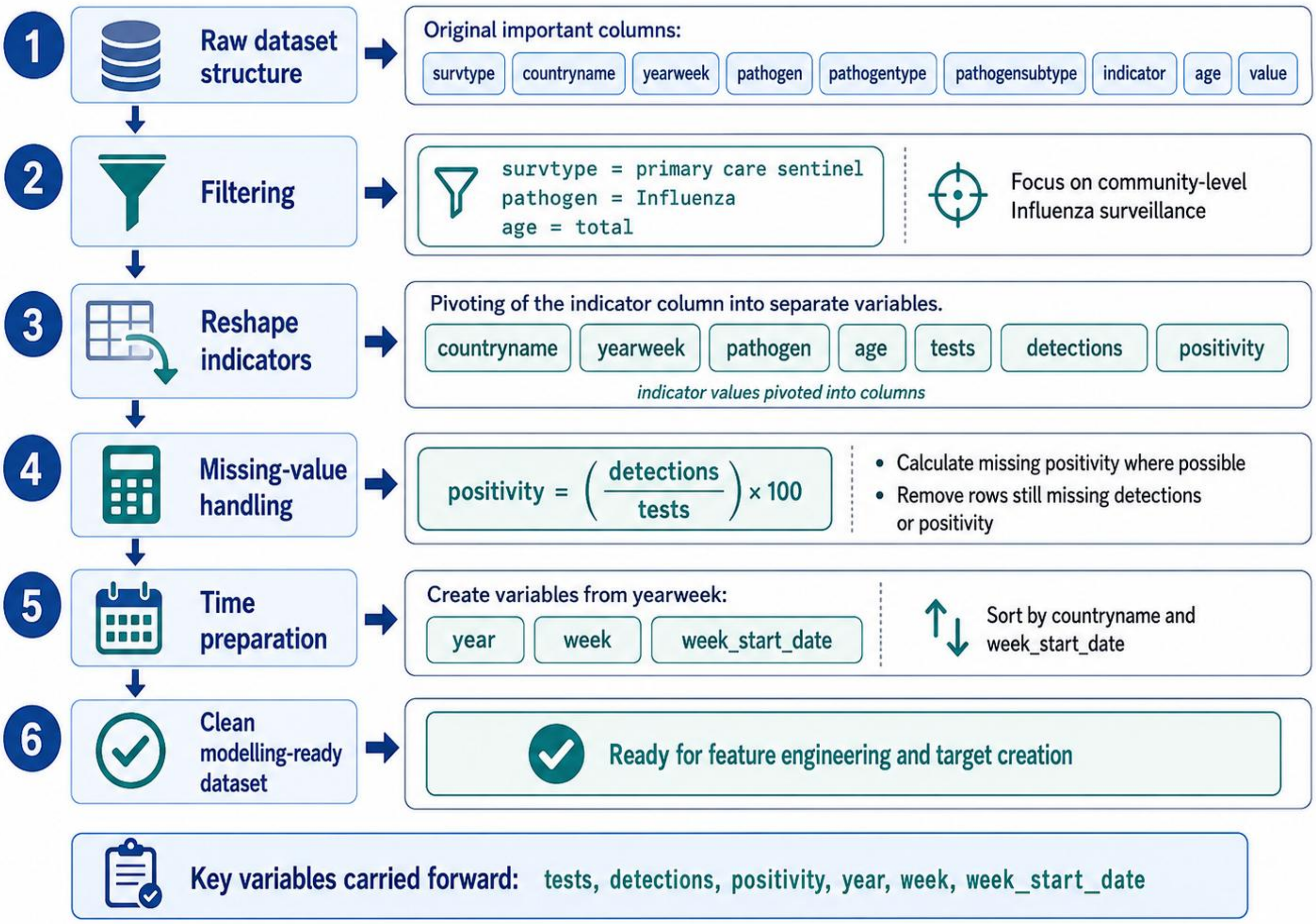


Figure 3.2.1: Data Cleaning and Reshaping Workflow

3.3 Data Preprocessing

Data preprocessing was a major part of the methodology because the raw surveillance data was not immediately ready for analysis. Several steps were required to transform it into a clean and machine-learning-friendly format.

The first step involved loading the dataset and inspecting its structure. This included checking the number of rows and columns, confirming the available variables, and identifying the unique values in key columns such as surveillance type, pathogen, indicator, and age.

The second step involved filtering the data to retain only **primary care sentinel** records for **Influenza**. This created the main analytical subset used for the study.

The third step involved reshaping the dataset. In the original format, the indicator type was stored in the indicator column, while the corresponding values were stored in the value column. The data was therefore pivoted so that the three main indicators — **tests**, **detections**, and **positivity** — became separate variables in the dataset. This created a much cleaner structure for both exploratory analysis and modelling.

The fourth step involved handling missing values. After reshaping, some rows contained missing values in the detections or positivity columns. Where possible, positivity was recalculated using test results and detections. Any rows still missing key modelling values after this step were removed to ensure that the final analytical dataset contained complete core indicators.

The fifth step involved creating time variables from the yearweek column. The year and week numbers were extracted, and a proper week_start_date variable was created to represent the weekly reporting period in chronological order. This was necessary for sorting the data correctly and for later time-based splitting and feature engineering.

3.4 Exploratory Data Analysis Plan

Before model building, exploratory data analysis was conducted to understand the behaviour of the cleaned Influenza dataset. This stage focused on identifying broad patterns in key surveillance indicators and on understanding how they changed over time and across reporting settings.

The exploratory analysis examined:

- the overall size and structure of the cleaned dataset
- the descriptive summary of tests, detections, and positivity
- weekly trends in positivity
- weekly trends in tests and detections
- country-level differences in Influenza activity
- distribution patterns of the main variables
- time-based and seasonal behaviour
- the role of lag and rolling features

The purpose of this stage was not only to describe the dataset, but also to help justify the feature engineering and modelling approach.

3.5 Feature Engineering

Feature engineering was one of the most important stages of the methodology. The goal of this stage was to create modelling features that captured not only the current surveillance situation, but also the recent short-term movement of Influenza activity.

The first group of features included the core current-week variables:

- tests
- detections
- positivity

The second group included **time-based variables**:

- year
- week
- week_start_date

These variables were useful because Influenza activity often follows seasonal patterns, and the timing of the reporting week may influence outbreak behaviour.

The third group included **lag features**:

- positivity_lag1
- detections_lag1
- tests_lag1

These variables captured the values from the previous week within each country. This was important because outbreak behaviour often depends on recent trajectory rather than current value alone.

The fourth group included **change-based features**:

- positivity_change
- detections_change

These variables captured the week-to-week increase or decrease in surveillance activity. They were useful because rising positivity or rising detections may act as early signs of increasing outbreak risk.

The fifth group included **rolling average features**:

- positivity_roll3
- detections_roll3
- tests_roll3

These were calculated as 3-week rolling averages within each country. The purpose of these features was to smooth short-term fluctuations and capture the broader recent trend more clearly.

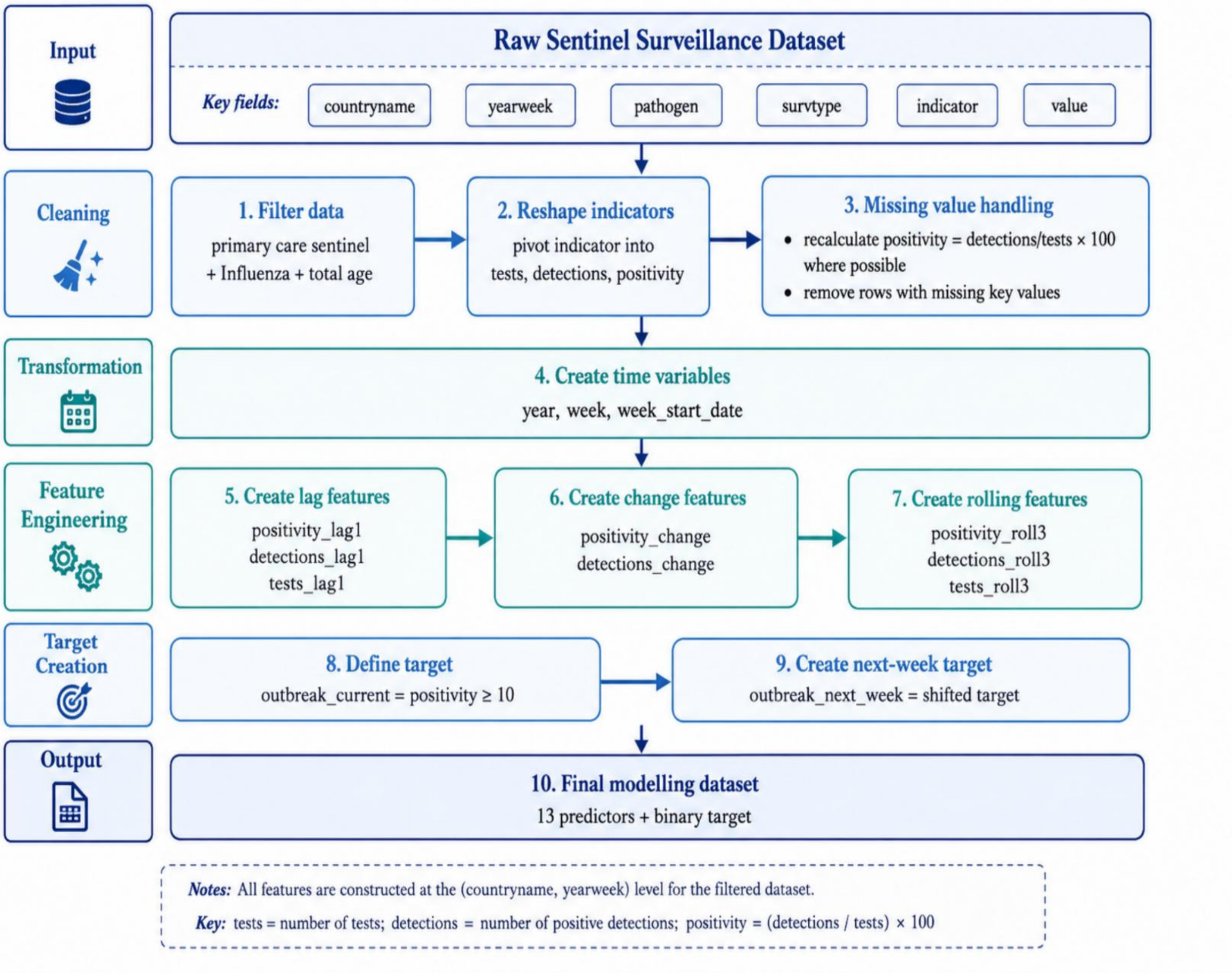


Figure 3.5.1: Data Preparation and Feature Engineering Pipeline

3.6 Definition of the Target Variable

The target variable used in the study was the next-week outbreak risk. The modelling task was therefore a binary classification problem in which the model predicted whether an outbreak would occur the following week.

To construct this target, the study first defined a current outbreak flag using positivity. A week was labelled as an outbreak week when the positivity value was greater than or equal to 10. Weeks below this threshold were labelled as non-outbreak weeks. This created a binary variable representing the current week's outbreak status.

The current outbreak flag was then shifted forward by 1 week within each country to create the final target variable, outbreak_next_week. This means the models were trained to learn from current and recent surveillance data to predict the outbreak status for the following week.

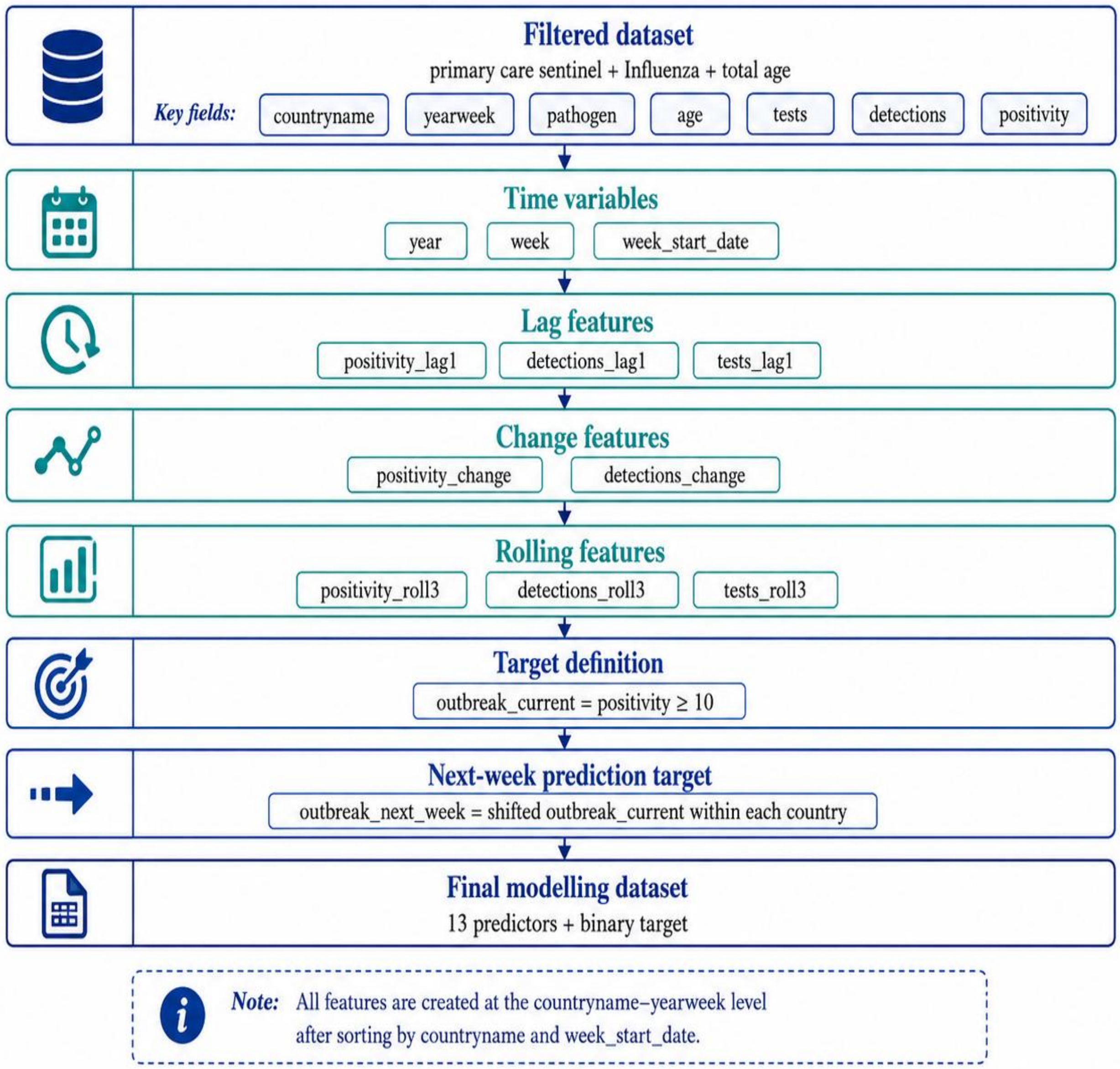


Figure 3.6.1: Feature Engineering and Target Creation Summary

3.7 Machine Learning Models Used

Five machine learning models were used in the dissertation:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- K-Nearest Neighbours

These models were chosen because they represent different modelling approaches and provide a useful balance between baseline interpretability and flexible predictive power.

Logistic Regression was used as a strong baseline classifier for binary outbreak prediction. It is simple, interpretable, and widely used in health-related classification tasks.

Decision Tree was used because it creates rule-based splits and can capture non-linear relationships in the data. It is also relatively easy to interpret compared with some other methods.

Random Forest was included because it combines multiple decision trees and generally performs strongly on structured tabular data. It is especially useful when the data contains more complex relationships that a single decision tree may not capture reliably.

Support Vector Machine was included because it is a strong classifier for structured numeric data and can model more complex decision boundaries. Since it is sensitive to variable scale, it was used within a pipeline that included feature standardisation.

K-Nearest Neighbours was included because it provides a distance-based classification approach. It predicts outbreak status based on similarity to nearby observations in the feature space. Like SVM, it also required feature standardisation.

3.8 Hyperparameter Tuning and Cross-Validation Strategy

A key methodological improvement in the final version of the study was the inclusion of hyperparameter tuning and time-series cross-validation. These steps were used to improve model robustness and ensure that the selected models were not based only on default parameter settings.

The first stage of validation involved working only within the training dataset. The training data was sorted chronologically and then divided into time-based cross-validation splits using TimeSeriesSplit with 5 splits. Rather than creating random validation folds, the cross-validation process was based on full weekly periods, which preserved the temporal structure of the outbreak data. This was important because the study focuses on time-dependent surveillance data and future prediction.

The tuning metric used during model selection was F1-score. This was an appropriate choice because the outbreak prediction problem requires a balance between precision and recall. A model that identifies many outbreak weeks but produces too many false alarms is not ideal, and a model that avoids false alarms but misses too many outbreaks is also not ideal. F1-score, therefore, provided a balanced optimisation target during tuning.

Two different search methods were used:

- **GridSearchCV** was used for:
    - Logistic Regression
    - K-Nearest Neighbours
- **RandomizedSearchCV** was used for:
    - Decision Tree
    - Random Forest
    - Support Vector Machine
- This strategy was practical because some models had smaller, more manageable search spaces, while others had larger parameter spaces where randomised search was more efficient.
- For **Logistic Regression**, tuning focused on the regularisation strength and class weighting.
- For **KNN**, tuning focused on the number of neighbours, distance weighting, and distance metric.
- For the **Decision Tree**, tuning focused on the split criterion, tree depth, minimum split size, minimum leaf size, and class weighting.
- For **Random Forest**, tuning focused on the number of trees, tree depth, split size, leaf size, feature sampling strategy, and class weighting.
- For **SVM**, tuning focused on the regularisation parameter, kernel type, gamma value, and class weighting.

For models that require scaling, a **pipeline** structure was used to ensure scaling and modelling were correctly performed during the search process. This applied to Logistic Regression, SVM, and KNN.

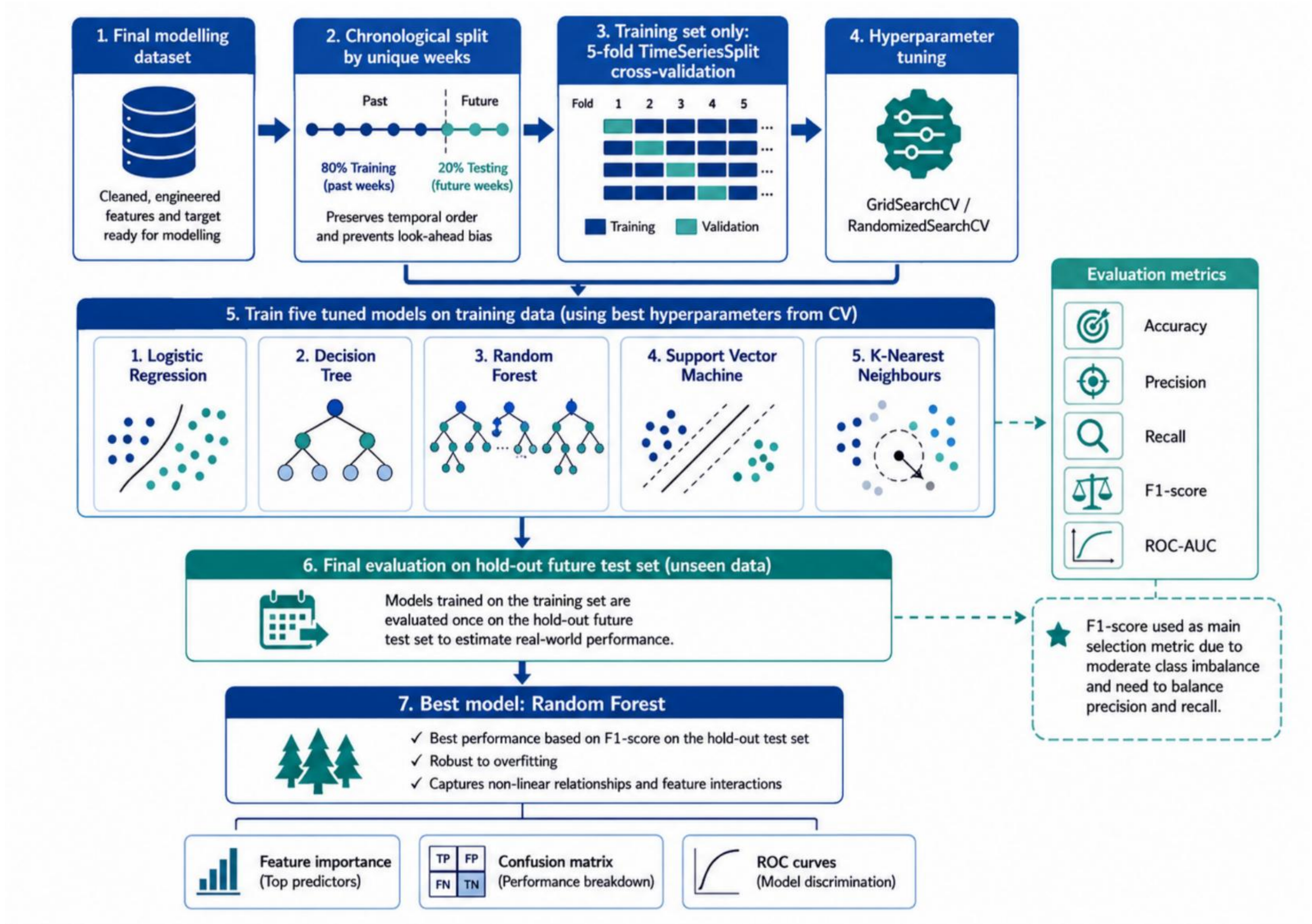


Figure 3.8.1: Machine Learning Development and Evaluation Workflow

3.9 Data Splitting and Final Evaluation Strategy

After the feature engineering stage, the final modelling dataset was split into training and testing periods using a chronological approach. The split was based on complete weekly periods rather than individual rows. This ensured that all observations from the same reporting week remained together in either the training set or the testing set.

The first 80% of unique weekly periods were used for training, and the last 20% for testing. This was a more realistic design than a random split because the study aimed to predict future outbreak status from past surveillance data.

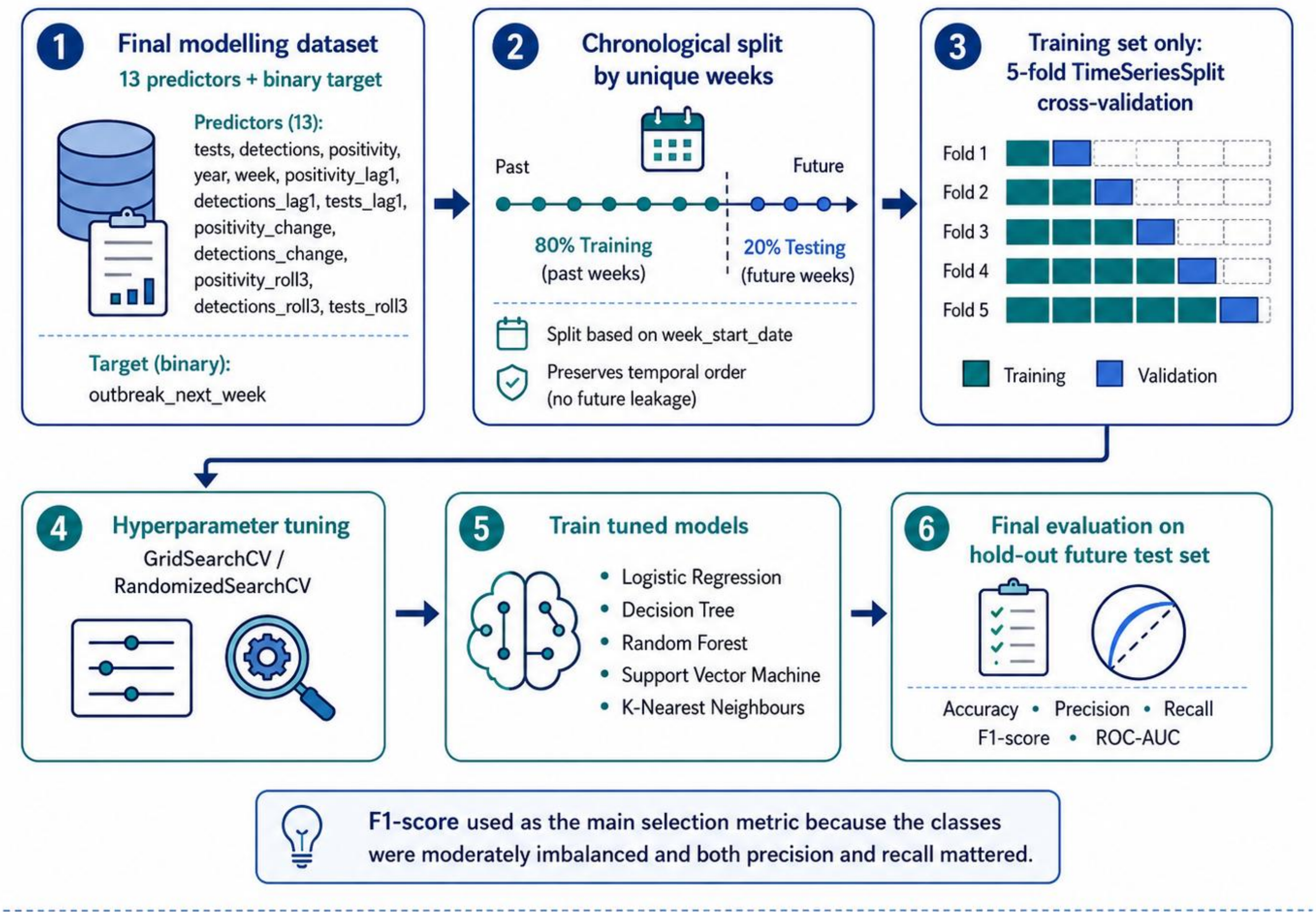


Figure 3.9.1: Time-Series Data Splitting and Validation Strategy

The training period covered the earlier part of the surveillance timeline, while the testing period contained later weeks that were not used at any stage of tuning or model fitting. This means the final test set acted as a true hold-out set for evaluating model performance.

This evaluation design had two major strengths:

1. The models were tuned using only the training set with time-series cross-validation.
2. The final performance was then assessed on a separate future test set.

This strengthened the methodology and made it more defensible. It reduced the risk of overly optimistic model estimates and ensured that the final reported performance reflected a more realistic outbreak prediction setting.

3.10 Model Evaluation Metrics

The final tuned models were evaluated on the hold-out test set using multiple classification metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Accuracy measures the overall proportion of correct predictions. While it is useful as a general summary, it is not sufficient on its own in an outbreak setting.

Precision measures how many predicted outbreak weeks were actually true outbreak weeks. This is useful for understanding false outbreak alarms.

Recall measures how many true outbreak weeks were successfully identified by the model. This is especially important in a public health context because missed outbreak weeks may reduce the usefulness of an early-warning framework.

F1-score combines precision and recall into one balanced metric. This was especially important in the present study and was also used as the tuning metric during cross-validation.

ROC-AUC measures how well the model separates outbreak and non-outbreak cases across possible thresholds. This provides a broader view of classification quality beyond one single cut-off.

3.11 Tools and Software Used

The study was carried out in **Python** using **Google Colab** as the main coding environment.

The main Python libraries used in the study included:

- **Pandas** for data handling and reshaping
- **NumPy** for numerical operations
- **Matplotlib** for plotting and visualisation
- **Scikit-learn** for machine learning models, pipelines, cross-validation, hyperparameter tuning, and evaluation metrics

CHAPTER 4: EXPLORATORY DATA ANALYSIS AND FINDINGS

This chapter presents the exploratory data analysis of the cleaned Influenza sentinel surveillance dataset used in the study. The main purpose of this chapter is to examine the structure and behaviour of the data before discussing the machine learning modelling results in the following chapter. Exploratory data analysis is an important stage in a data-driven study because it helps reveal patterns, relationships, and irregularities within the dataset.

4.1 Overview of the Cleaned Dataset

The final cleaned dataset used in this study was derived from the primary care sentinel surveillance dataset after filtering for Influenza and applying all preprocessing steps described in Chapter 3. These preprocessing steps included filtering the relevant records, reshaping the data into a column-based format for tests, detections, and positivity, handling missing values, generating time variables, and creating lag and rolling features for modelling purposes.

After cleaning, the main Influenza dataset contained **4,584 observations**, with each row representing an aggregated surveillance record for a given country and reporting week. The dataset covered a multi-country European surveillance context and included observations across a broad weekly time span. The main variables used in the exploratory analysis were:

- number of tests,
- number of detections,
- positivity rate,
- year,
- week,
- lag variables,
- change variables,
- and rolling average variables.

The cleaned dataset had no missing values in the key analytical variables required for the study, namely tests, detections, and positivity.

4.2 Descriptive Summary of Key Variables

The three most important variables in this study are tests, detections, and positivity. Together, these variables provide a broad view of how often testing was carried out, how many Influenza detections were reported, and what proportion of tested samples were positive during a given week.

The variable **tests** represent the number of samples tested within the sentinel surveillance system. This variable is important because it reflects surveillance effort and helps place the detection counts in context. A high number of detections does not necessarily indicate a strong outbreak if the number of tests is also very high. Therefore, tests are useful but should not be interpreted alone.

The variable **detections** represent the number of confirmed Influenza-positive samples. This variable provides a more direct measure of virus activity than tests alone, but it still depends partly on how much testing was carried out. For this reason, detections are informative, but they are often better interpreted together with positivity.

The variable **positivity** represents the percentage of tested samples that were positive for Influenza. This was one of the most important variables in the study because it offers a more proportion-based measure of virus circulation. A rise in positivity suggests that a larger share of tested individuals is testing positive, which often gives a clearer signal of increasing activity than tests or detections alone.

4.3 Weekly Trend of Influenza Positivity

One of the most important parts of the exploratory analysis was the examination of positivity over time. Positivity is a particularly valuable surveillance indicator because it captures the proportion of tested samples that were positive for Influenza in each reporting week. Unlike raw detection counts, it allows the researcher to account for differences in testing activity and focus more directly on the intensity of virus circulation.

The weekly positivity trend showed that Influenza activity did not remain constant across the study period. Instead, it changed over time, with some weeks showing relatively low positivity and other periods showing clear increases.

4.4 Weekly Trend of Tests and Detections

In addition to positivity, the dataset was explored using the weekly trends of tests and detections. These two indicators are important because they provide a broader view of surveillance activity and infection detection across time.

The weekly trend in **tests** reflects the level of surveillance effort or testing activity in a given week. This variable showed that testing was not identical across all weeks. At some points, testing levels were higher, while at others they were lower. This variation is expected in real-world surveillance systems, where testing intensity may vary with seasonal conditions, healthcare demand, and reporting practices.

The weekly trend in **detections** provides a more direct indication of the number of Influenza-positive samples identified during each week. In general, detections tend to become more meaningful when interpreted together with positivity. A rise in detections can indicate increasing virus activity, but this interpretation is stronger when positivity rises at the same time. If detections increase only because testing increases, the outbreak implication may be weaker. However, if detections and positivity rise together, then the signal becomes more convincing.

4.5 Country-Level Comparison

The country-level analysis suggested that Influenza activity was not distributed evenly across all countries. Some countries appeared to show stronger positivity values, larger detection counts, or more visible outbreak peaks than others. This is not surprising, as surveillance systems may differ in scale, reporting coverage, healthcare structure, and the timing of seasonal virus circulation.

4.6 Distribution Analysis of Key Variables

The distribution of **positivity** showed that many observations were concentrated at relatively lower levels, with fewer weeks showing clearly elevated positivity values. This kind of pattern is expected in respiratory virus surveillance because outbreak periods usually occur only during parts of the year, while many other weeks show lower activity. Therefore, the dataset includes both routine lower-intensity weeks and smaller numbers of higher-risk weeks.

The distribution of **detections** showed a similar pattern. In many observations, the number of detections remained at lower levels, while some weeks displayed much stronger activity. This again reflects the fact that Influenza activity is not constant across time.

The distribution of **tests** may show a wider spread, since testing practices can vary across countries and weeks. Testing counts are useful for context, but the distribution also reminds us that direct count variables should not be interpreted without considering the broader surveillance situation.

4.7 Time-Based and Seasonal Pattern Analysis

Time and seasonality are central to respiratory virus surveillance. Influenza does not occur randomly across the calendar. Instead, it tends to show stronger activity during certain parts of the year.

The weekly structure of the data showed that the **week** variable was meaningful in understanding the behaviour of Influenza activity. This was later confirmed in the machine learning stage, where the **week** variable appeared among the more important predictors in the Random Forest feature importance results. This suggests that the timing of the year carries useful information about when outbreak risk is more likely to rise.

Seasonal pattern analysis is important because it provides a practical explanation for why outbreak risk is not uniform across all weeks. Influenza activity tends to rise in more concentrated seasonal windows and fall during other parts of the year. Even without using highly advanced seasonal modelling, this pattern can still be captured by calendar-based variables such as week number and rolling positivity behaviour.

CHAPTER 5: MACHINE LEARNING MODELLING AND EVALUATION

This chapter presents the machine learning modelling results of the study. After the exploratory analysis described in Chapter 4, the next stage of the research was to build and evaluate models capable of predicting whether an Influenza outbreak would occur in the following week. The aim of this stage was not only to build a predictive model, but also to compare multiple machine learning approaches and identify the one that performed best on the cleaned sentinel surveillance dataset.

5.1 Logistic Regression Results

The first tuned model applied in the study was Logistic Regression. Logistic Regression was used as an important baseline classifier because it is one of the most widely used methods for binary classification and remains relatively easy to interpret.

For Logistic Regression, hyperparameter tuning was carried out using **GridSearchCV** within the training dataset. The search process used time-series cross-validation and F1-score as the selection metric.

The best **cross-validation F1-score** for the tuned Logistic Regression model was **0.7770**. This indicates that even during internal validation, the model was already showing good predictive potential.

When evaluated on the hold-out test set, the tuned Logistic Regression achieved:

```
Logistic Regression Performance:
Accuracy  : 0.8646
Precision : 0.8697
Recall    : 0.7976
F1-Score  : 0.8321
ROC-AUC   : 0.9181

====================================
Confusion Matrix:
[[516  49]
 [ 83 327]]
```

Figure 5.1.1: Logistic Regression Performance Results

The confusion matrix for Logistic Regression showed that the model correctly classified **516 non-outbreak weeks** and **327 outbreak weeks**, while producing relatively limited false alarms and missed outbreaks. This made it a strong tuned baseline for the rest of the comparison.

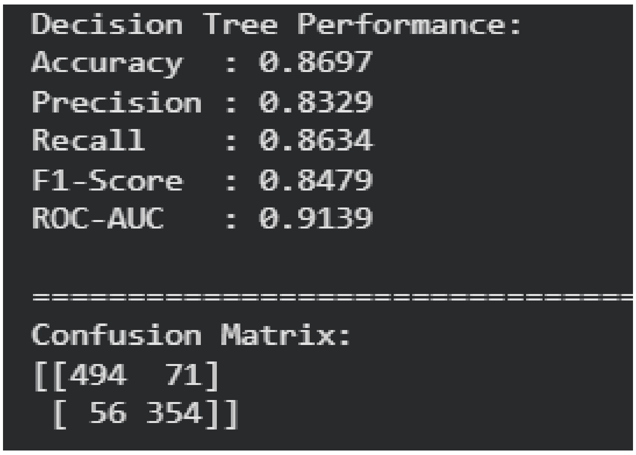
5.2 Decision Tree Results

The second tuned model applied in the study was Decision Tree. Decision Tree was included because it can capture non-linear relationships and produces rule-based classification logic. Unlike Logistic Regression, it does not rely on a linear boundary and can adapt more flexibly to different decision patterns within the feature space.

For the Decision Tree model, **RandomizedSearchCV** was used together with time-series cross-validation and F1-score optimisation.

The best **cross-validation F1-score** for the tuned Decision Tree model was **0.8005**, which was stronger than that of Logistic Regression in cross-validation.

When evaluated on the hold-out test set, the tuned Decision Tree achieved:



```
Decision Tree Performance:
Accuracy  : 0.8697
Precision : 0.8329
Recall    : 0.8634
F1-Score  : 0.8479
ROC-AUC   : 0.9139

================================
Confusion Matrix:
[[494  71]
 [ 56 354]]
```

Figure 5.2.1: Decision Tree Performance Results

The confusion matrix showed that the model correctly identified **494 non-outbreak weeks** and **354 outbreak weeks**. This demonstrates that the tuned Decision Tree was especially strong in capturing true outbreak periods, although it did so with slightly more false positives.

5.3 Random Forest Results

The Random Forest model was the strongest and most important model in the study. Random Forest is an ensemble learning method that combines many decision trees in order to improve generalisation and reduce instability. It is especially well suited to structured surveillance data because it can capture complex non-linear patterns while remaining more robust than a single decision tree.

For Random Forest, **RandomizedSearchCV** was used within the training data together with time-series cross-validation and F1-score optimisation.

The best **cross-validation F1-score** for the tuned Random Forest model was **0.8249**, which was the highest among all five models. This was already a strong indication that Random Forest would perform especially well in the final evaluation.

When tested on the chronological hold-out set, the tuned Random Forest model achieved:

```
Random Forest Performance:
Accuracy  : 0.9026
Precision : 0.9070
Recall    : 0.8561
F1-Score  : 0.8808
ROC-AUC   : 0.9444

==============================
Confusion Matrix:
[[529  36]
 [ 59 351]]
```

Figure 5.3.1: Random Forest Performance Results

The confusion matrix showed that the tuned Random Forest correctly classified **529 non-outbreak weeks** and **351 outbreak weeks**, while keeping false alarms and missed outbreaks relatively low. This gives the model strong practical value as an early-warning framework.

5.4 Support Vector Machine Results

The Support Vector Machine (SVM) model also performed strongly after tuning. SVM was included because it is a powerful classification method that can handle complex decision boundaries, especially when the feature values are standardised. Since the surveillance dataset contains multiple numeric variables measured on different scales, scaling was applied through a pipeline before tuning and model fitting. For SVM, **RandomizedSearchCV** was used together with time-series cross-validation.

The best **cross-validation F1-score** for the tuned SVM model was **0.7917**.

When evaluated on the hold-out test set, the tuned SVM achieved:

```
Support Vector Machine Performance:
Accuracy  : 0.8821
Precision : 0.8554
Recall    : 0.8659
F1-Score  : 0.8606
ROC-AUC   : 0.9307

======================================
Confusion Matrix:
[[505  60]
 [ 55 355]]
```

Figure 5.4.1: Support Vector Machine Performance Results

The confusion matrix showed that the model correctly classified **505 non-outbreak weeks** and **355 outbreak weeks**. This means it performed well on both classes and was particularly useful in identifying outbreak weeks.

5.5 K-Nearest Neighbours Results

The final tuned model included in the study was K-Nearest Neighbours (KNN). KNN is a distance-based classifier that predicts the class of a new observation based on the labels of the nearest similar observations in the feature space. Because KNN depends heavily on distance, feature standardisation was applied through a pipeline before tuning and model fitting.

For KNN, **GridSearchCV** was used together with time-series cross-validation and F1-score optimisation.

The best **cross-validation F1-score** for the tuned KNN model was **0.8036**, which was relatively strong and ranked among the better validation results across the models.

When evaluated on the final test set, the tuned KNN achieved:

```
K-Nearest Neighbours Performance:
Accuracy  : 0.8903
Precision : 0.8741
Recall    : 0.8634
F1-Score  : 0.8687
ROC-AUC   : 0.9200

======================================
Confusion Matrix:
[[514  51]
 [ 56 354]]
```

Figure 5.5.1: KNN Performance Results

The confusion matrix showed that KNN correctly classified **514 non-outbreak weeks** and **354 outbreak weeks**. This confirms that it handled both outbreak and non-outbreak classes effectively.

5.6 Comparative Performance of All Models

The comparison of all five tuned machine learning models is one of the most important parts of this chapter. It allows the study to go beyond individual model interpretation and make a justified decision about which tuned model performed best for next-week Influenza outbreak prediction.

First, all five tuned models performed reasonably well, which confirms that the selected surveillance-based feature set was meaningful and that the tuning process improved model reliability.

Second, Random Forest emerged as the best overall model because it achieved:

- the highest **cross-validation F1-score**
- the highest **test F1-score**
- the highest **accuracy**
- and the highest **ROC-AUC**

This made Random Forest the strongest overall choice.

Final Model Comparison Table:

	Model	Best_CV_F1	Accuracy	Precision	Recall	F1_Score	ROC_AUC
0	Random Forest	0.8249	0.9026	0.9070	0.8561	0.8808	0.9444
1	K-Nearest Neighbours	0.8036	0.8903	0.8741	0.8634	0.8687	0.9200
2	Support Vector Machine	0.7917	0.8821	0.8554	0.8659	0.8606	0.9305
3	Decision Tree	0.8005	0.8697	0.8329	0.8634	0.8479	0.9139
4	Logistic Regression	0.7770	0.8646	0.8697	0.7976	0.8321	0.9181

```
============================================================
Best model based on F1-score:
Model          Random Forest
Best_CV_F1            0.8249
Accuracy              0.9026
Precision              0.907
Recall                0.8561
F1_Score              0.8808
ROC_AUC               0.9444
Name: 0, dtype: object
```

Figure 5.6.1: Final Model Comparison Table

Third, the tuned KNN model performed even more strongly than expected and became the second-best model overall in terms of F1-score. This is an important difference from the earlier untuned comparison and shows that tuning had a meaningful effect.

Fourth, SVM also remained strong, especially in recall and ROC-AUC, but its overall balance was still slightly below KNN and Random Forest. Decision Tree improved meaningfully after tuning and performed better than Logistic Regression in F1-score. Logistic Regression remained a strong baseline, but it became the weakest overall model among the five once tuning was fully applied across all models.

Based on this tuned comparison, Random Forest was selected as the final best-performing model in the dissertation.

5.7 Feature Importance and Model Interpretation

Once the tuned Random Forest was identified as the best model, the next step was to examine which variables contributed most to its predictions. This part of the analysis is especially important because it improves interpretability. A strong model is useful, but a strong model that can also show which features matter most is more meaningful in a public health setting.

These results are highly meaningful. They show that **current positivity** was by far the most important predictor of next-week outbreak risk. Its importance score was much higher than that of the other variables, which suggests that the current proportion of positive Influenza tests was the single strongest signal in the dataset.

The **week** variable was the second most important feature, which shows that seasonal timing remained highly relevant even in the tuned model. This supports the argument that Influenza activity depends not only on current surveillance levels, but also on the point of the seasonal cycle in which the observation occurs.

The importance of **positivity_roll3** also confirms that recent short-term average positivity mattered to the model. This means that not only the current positivity level, but also the broader recent positivity pattern helped shape next-week outbreak prediction.

Feature Importance from Tuned Random Forest:

	Feature	Importance
0	positivity	0.6943
1	week	0.0691
2	positivity_roll3	0.0555
3	detections_roll3	0.0516
4	tests_roll3	0.0199
5	tests_lag1	0.0195
6	tests	0.0191
7	positivity_lag1	0.0184
8	positivity_change	0.0163
9	year	0.0111
10	detections	0.0102
11	detections_change	0.0093
12	detections_lag1	0.0058

Figure 5.7.1: Feature Importance from Tuned Random Forest

5.8 Visual Evaluation of Model Performance

In addition to numerical performance metrics, the study also used visual methods to evaluate and explain the results of the tuned models.

The first important visual output was the **feature importance chart** of the tuned Random Forest model. This figure clearly showed that positivity was the dominant predictor, followed by week and recent rolling features. This visual output helped make the model more interpretable and easier to explain in the dissertation and viva.

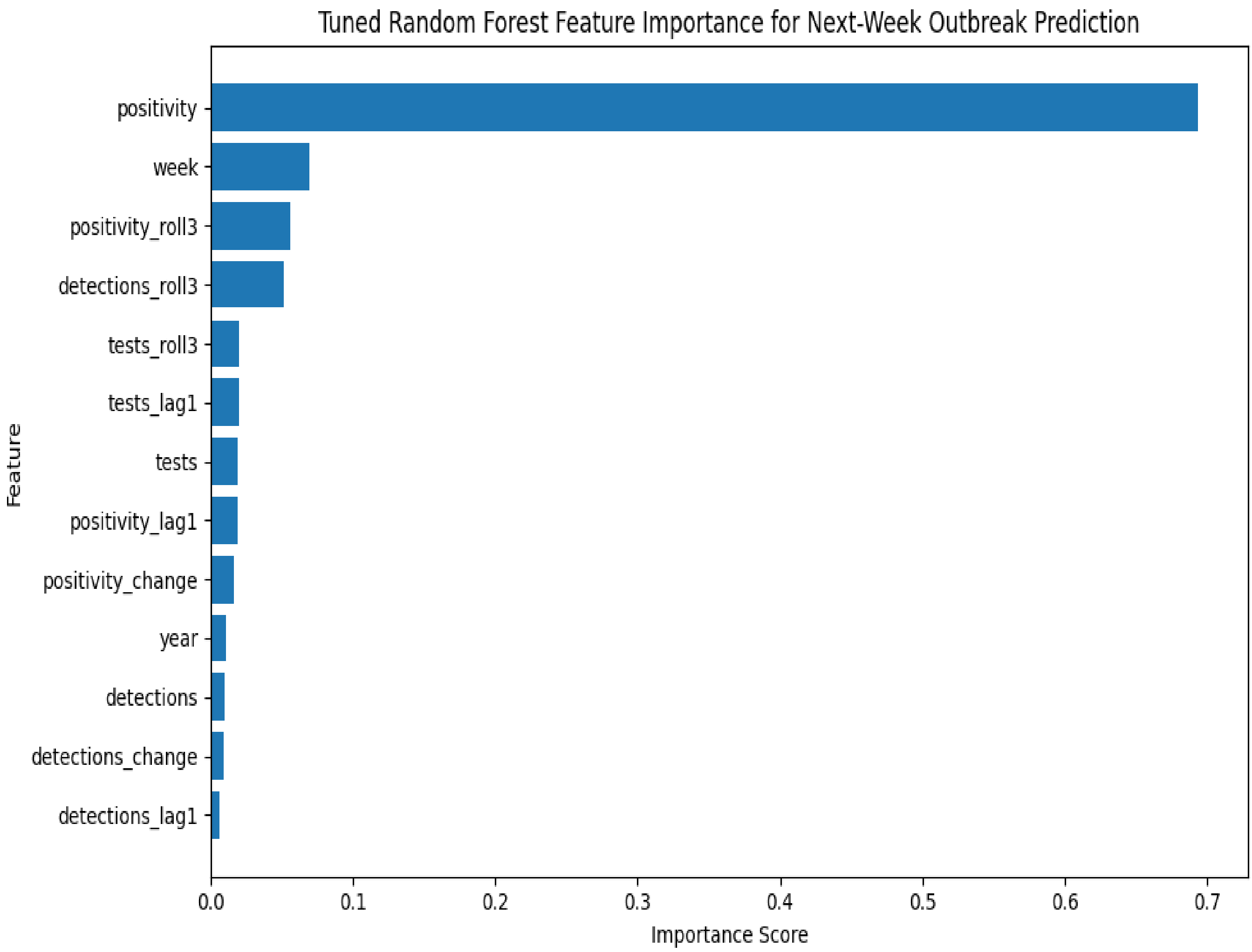


Figure 5.8.1: Feature importance chart

The second major visual output was the **confusion matrix** of the tuned Random Forest model. This provided a direct picture of how the best model classified outbreak and non-outbreak weeks. It showed that the model correctly classified most observations in both classes, while keeping false positives and false negatives relatively low.

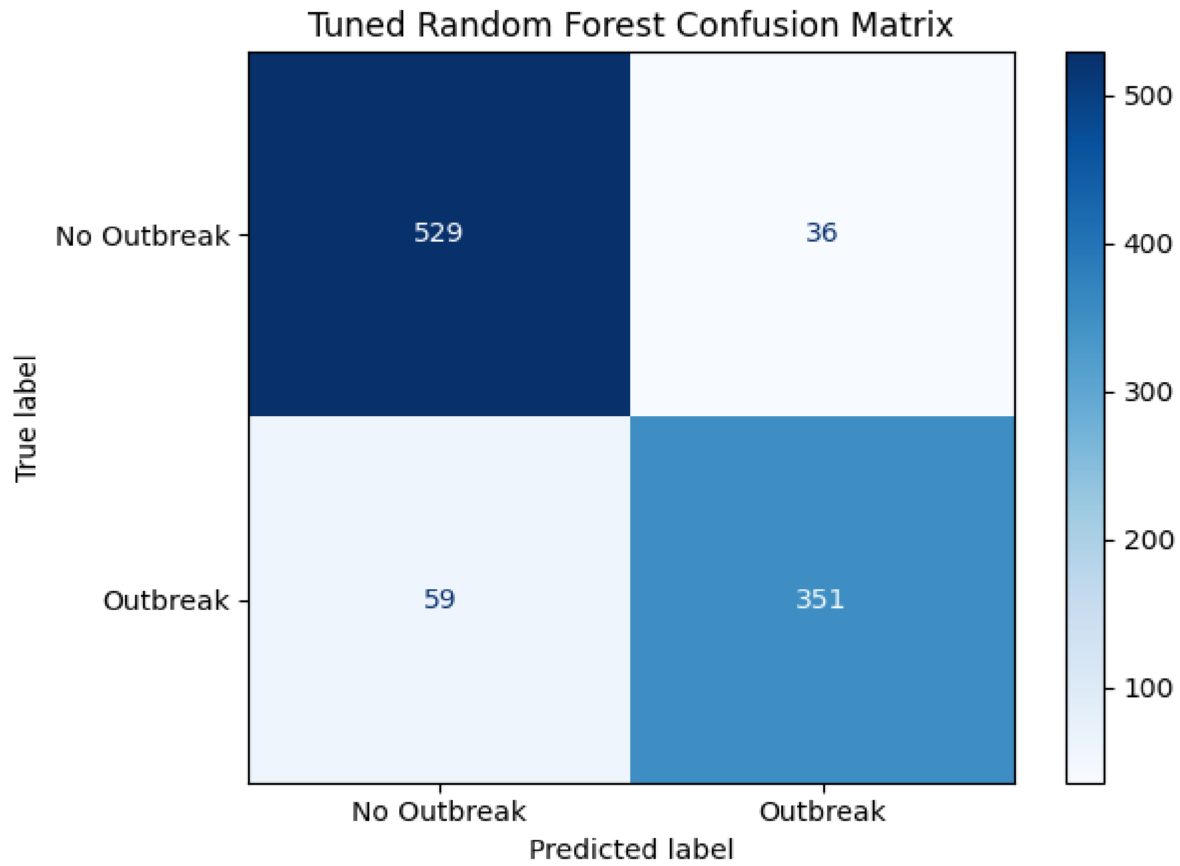


Figure 5.8.2: Confusion matrix

The third key visual output was the **ROC curve comparison** of all five tuned models. This figure visually showed the relative classification strength of the models across different thresholds and confirmed that Random Forest had the strongest ROC-AUC overall. It also showed that the other tuned models, especially KNN and SVM, remained competitive but still slightly behind the final selected model.

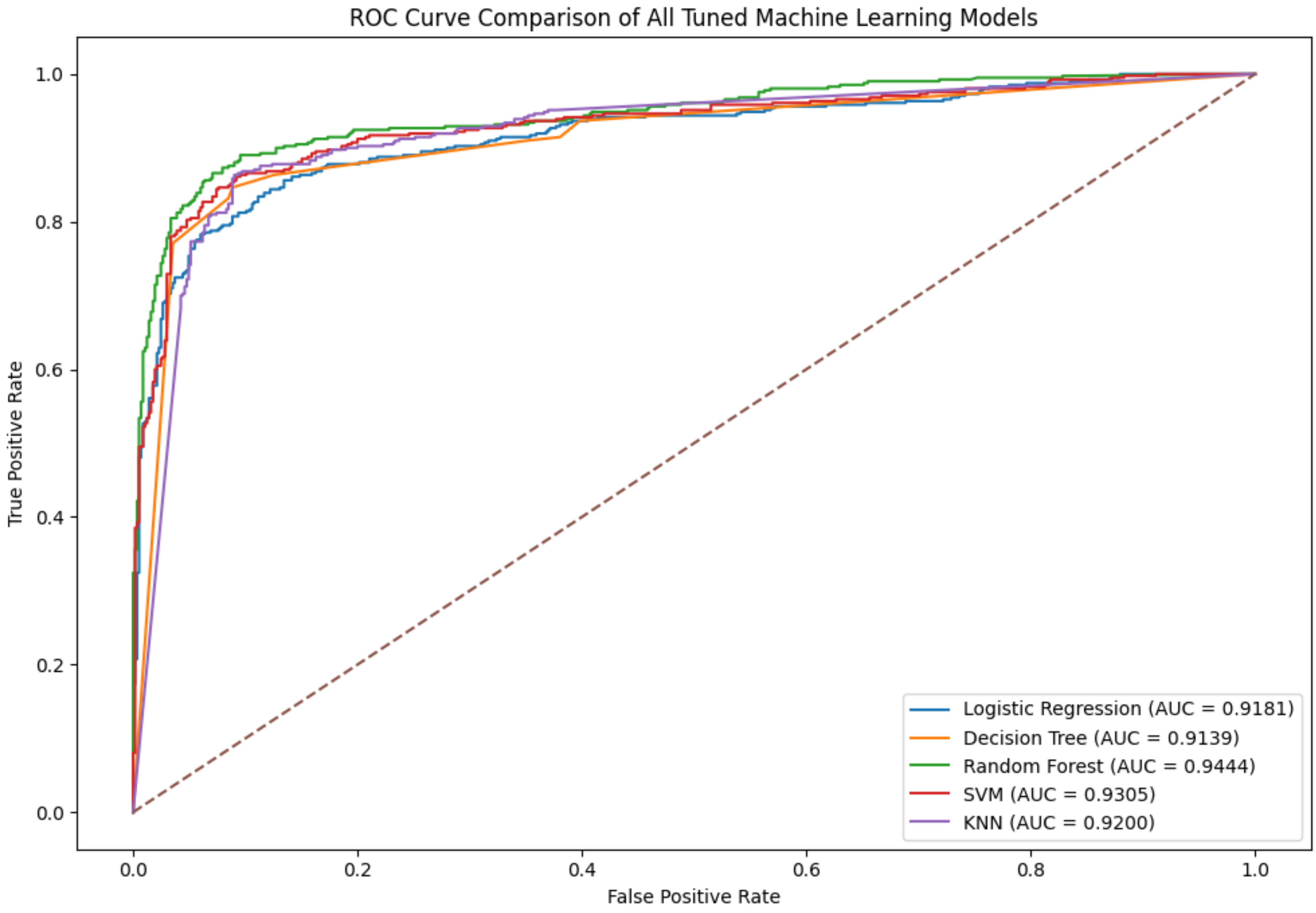


Figure 5.8.3: ROC curve comparison of all five tuned models

CHAPTER 6: DISCUSSION

This chapter discusses the main findings of the study and compares them with the literature reviewed in Chapter 2. The purpose of this chapter is not only to repeat the results, but to explain what they mean in relation to previous research on respiratory virus surveillance, early warning, and machine learning-based outbreak prediction.

6.1 Discussion of Machine Learning Results

The machine learning results showed that next week's influenza outbreak prediction was possible using official public health surveillance data. This finding agrees with earlier studies, which showed that surveillance data can be used not only for monitoring past activity but also for short-term prediction and early warning.

All five tuned models in this dissertation performed reasonably well, which suggests that the final feature set contained meaningful predictive information. This supports the findings of previous forecasting studies, such as Reich et al. (2019), which showed that model comparison and ensemble approaches are important for influenza forecasting. In the present study, different models showed varying strengths, confirming that model choice matters. Logistic Regression provided a strong baseline; Decision Tree offered rule-based classification; SVM and KNN performed strongly after scaling and tuning; and Random Forest produced the best overall performance.

The tuned **Random Forest** model achieved the strongest results in this study. This aligns with Cheng et al. (2020), who included Random Forests as part of an Influenza forecasting framework and showed that machine learning models can perform well when applied to structured surveillance data. It also aligns with Yang et al. (2024), who used Random Forest as one of the comparison models alongside a deep learning early-warning model. Although the present study did not use deep learning, it showed that a tuned Random Forest can still provide strong and interpretable performance for next-week outbreak prediction.

The strong performance of K-Nearest Neighbours and Support Vector Machine also supports the broader literature. Santillana et al. (2016) used a Support Vector Machine-based framework for real-time Influenza surveillance estimation, showing that margin-based models can be useful in this area. In the present study, SVM also performed strongly, especially in terms of ROC-AUC, although it did not outperform Random Forest. KNN also performed well, which suggests that weeks with similar surveillance patterns may have similar outbreak outcomes.

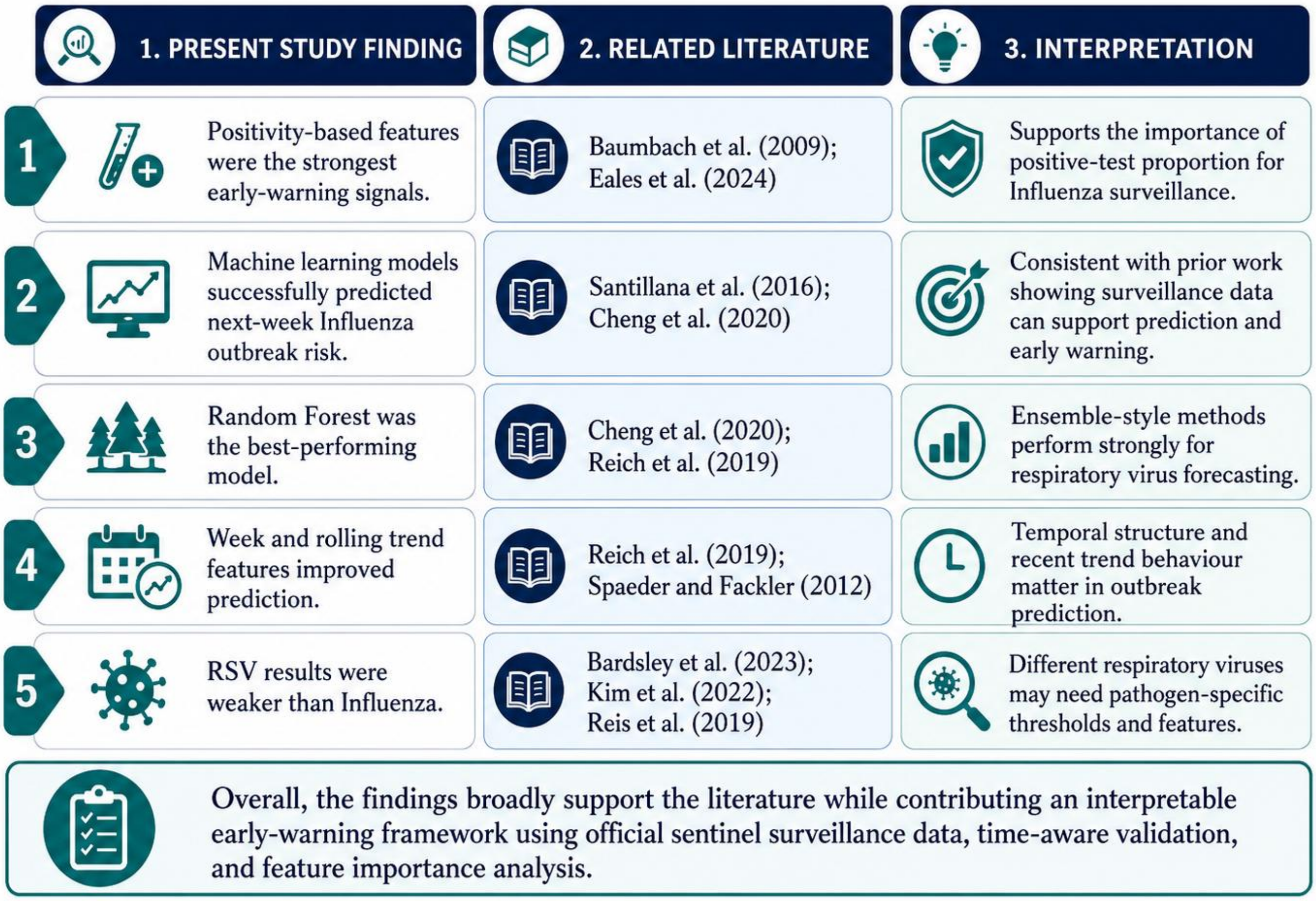


Figure 6.1.1: Discussion Summary and Literature Comparison

6.2 Why Random Forest Performed Best

The tuned Random Forest model performed best because it was well-suited to the dataset's structure and the type of prediction problem. The dataset contained current surveillance values, lag features, rolling averages, and time-based variables. These variables can interact in non-linear ways.

This result agrees with the wider machine learning literature reviewed in Chapter 2. Cheng et al. (2020) used Random Forests alongside ARIMA, Support Vector Regression, and Extreme Gradient Boosting in a forecasting framework and found that machine learning models were useful for Influenza prediction. Reich et al. (2019) also showed that ensemble-based approaches can perform strongly in seasonal Influenza forecasting. Although Random Forest is not a multi-model ensemble, it is still an ensemble model because it combines many decision trees.

The result also makes practical sense when compared with studies on surveillance indicators. Baumbach et al. (2009) found that positive rapid Influenza test results could rise earlier than traditional influenza-like illness indicators. This supports the present study's finding that positivity was a very strong predictor. Random Forest performed well because it could combine current positivity with lag and rolling features, rather than treating each variable in isolation.

6.3 Answering the Research Questions

Research Question 1: What temporal and surveillance patterns are associated with rising Influenza activity in public health surveillance data?

The study found that rising Influenza activity was mainly associated with increases in positivity, changes in detections, rolling positivity behaviour, and seasonal week position. This finding agrees with Baumbach et al. (2009), who showed that positivity-related indicators could provide earlier signals of Influenza activity than broader influenza-like illness measures. It also connects with Eales et al. (2024), who explained that routine surveillance indicators such as test-positive proportion are useful but must be interpreted carefully because they can be influenced by testing behaviour and other factors.

The present study supports the idea that temporal behaviour matters. The use of lag and rolling features showed that outbreak risk was not only linked to the current week, but also to recent short-term movement. This is consistent with forecasting studies such as Spaeder and Fackler (2012), Reis et al. (2019), and Reich et al. (2019), where time-based patterns were central to respiratory virus prediction. Therefore, the findings confirm that rising Influenza activity is best understood through both current surveillance values and recent time-based trends.

Research Question 2: Which surveillance indicators are most useful for identifying early outbreak signals?

The most useful indicators in this study were positivity-based variables, especially current positivity and rolling positivity. The final tuned Random Forest model showed that current positivity was the strongest feature, followed by week and recent rolling features. This strongly agrees with Baumbach et al. (2009), who found that the proportion of positive rapid Influenza tests increased earlier than influenza-like illness visit percentages across multiple seasons. It also supports the argument that positivity can be a clearer signal than raw counts because it considers the proportion of tests that are positive.

However, the findings also agree with Eales et al. (2024), who warned that positivity and other routine surveillance indicators are not perfect measures of true infection incidence. In the present study, positivity was very useful, but it was not interpreted alone. Tests, detections, lag features, and rolling averages were also included to provide broader surveillance context. This makes the framework more balanced than a system based only on one indicator.

Research Question 3: Which machine learning model performs best for predicting next-week Influenza outbreak risk from surveillance data?

Among the five tuned machine learning models tested, the tuned Random Forest model performed best overall. It achieved the highest cross-validation F1-score, final F1-score, accuracy, and ROC-AUC. This result is consistent with earlier literature showing that flexible and ensemble-style models are often strong for respiratory virus forecasting. Cheng et al. (2020) found that machine learning models, including Random Forest and ensemble methods, were effective for real-time Influenza forecasting. Reich et al. (2019) also showed that ensemble forecasting approaches performed strongly in seasonal Influenza prediction.

The result differs slightly from Yang et al. (2024), where a deep learning model outperformed traditional machine learning models such as Logistic Regression, SVM, Random Forest, and XGBoost. However, this difference is understandable because the present dissertation focused on a simpler, more interpretable machine learning framework rather than a deep learning system. For the scope of this study, Random Forest was more suitable because it provided both strong predictive performance and interpretable feature importance.

Therefore, the best-performing algorithm in this dissertation was Random Forest. It offered the strongest balance of accuracy, F1-score, ROC-AUC, and interpretability.

Research Question 4: How interpretable and practically useful are the model outputs for public health monitoring and decision-making?

The model outputs were useful and interpretable because they did not only produce performance scores, but also showed which variables were most important. The feature importance analysis showed that positivity, week, and rolling positivity features were central to the model's predictions. This is important because public health users need to understand why a model gives an outbreak warning, not only whether the prediction is correct.

This finding links directly with the research gap identified in Chapter 2. Some previous studies focused mainly on forecasting accuracy or complex modelling. For example, Yang et al. (2023) used Baidu search data and deep learning methods, while Yang et al. (2024) used a deep learning early-warning model. These approaches can perform strongly, but they may be harder to explain in simple public health terms. In contrast, the present study used a model that still performed strongly while allowing interpretation through feature importance.

The practical value of this study is also supported by Hammond et al. (2022), who argued that traditional surveillance systems remain important and that alternative data sources should support rather than replace them. The present dissertation used official sentinel surveillance data, which makes the framework more directly connected to routine public health monitoring. Maroufi et al. (2025) also highlighted that not every surveillance system is equally suitable for forecasting. This supports the present study's decision to focus on primary care sentinel Influenza data, which was more appropriate for early-warning analysis.

6.4 Brief Note on RSV as a Secondary Exploratory Case

Although the dissertation focused mainly on Influenza, RSV was considered as a secondary exploratory case. However, the predictive performance for RSV was weaker than the Influenza results, especially in terms of F1-score and recall. This suggests that the same modelling workflow may not transfer equally well across different respiratory viruses.

This finding is supported by earlier RSV-related studies. Bardsley et al. (2023) showed that RSV patterns changed strongly during and after the COVID-19 pandemic, including an unusual summer surge after a period of very low activity. Kim et al. (2022) also showed that RSV and Influenza behaved differently during the pandemic period, with RSV activity increasing while Influenza activity remained very low. These studies support the idea that different respiratory viruses can behave differently even under the same broad surveillance conditions.

The weaker RSV performance in this dissertation therefore does not mean that RSV cannot be predicted. Instead, it suggests that RSV may need a more pathogen-specific modelling approach. For example, RSV may require a different outbreak threshold, different seasonal assumptions, or different feature engineering. This is also consistent with Reis et al. (2019), who showed that RSV forecasting can vary across national, regional, and state levels, meaning that RSV prediction may be sensitive to geography and outbreak timing.

CHAPTER 7: LIMITATIONS AND ETHICAL CONSIDERATIONS

This chapter discusses the main limitations and ethical considerations of the study. It explains the data-related, methodological, and model-related limitations of the dissertation and briefly reflects on RSV as a secondary exploratory case. It also outlines the ethical issues linked to the use of surveillance data and predictive modelling, and highlights the main strengths of the study. The purpose of this chapter is to present the research in a balanced, realistic, and academically responsible way.

7.1 Data-Related Limitations

One of the main limitations of the study is related to the nature of the dataset itself. The research used **aggregated public health surveillance data**, which is very useful for monitoring overall disease activity, but does not provide the same level of detail as patient-level clinical data. This means that the analysis was carried out at the level of country-week surveillance records rather than at the level of individual patients. As a result, the study can identify broad patterns in Influenza activity, but it cannot explain how outbreak risk varies according to individual demographic or clinical factors.

The dataset also required cleaning before modelling could begin. Missing values were present in some country-week records before preprocessing, especially in the detections and positivity fields. These rows were handled carefully, but the fact that such missingness existed is itself a limitation. It suggests that surveillance reporting was not perfectly complete in all cases. Although the final modelling dataset was cleaned and contained no missing values in the main variables, the earlier data quality issues should still be acknowledged.

Another data-related limitation is that surveillance indicators represent **observed reported activity**, not the full true burden of infection in the community. Not all infections are tested, not all cases enter the healthcare system, and not all countries may apply identical surveillance intensity across time. Therefore, the dataset should be seen as a measure of surveillance-recorded virus activity rather than a complete measure of all Influenza infections in the population.

7.2 Methodological Limitations

The methodological scope of the study was also intentionally focused on **Influenza** as the main modelling target. This helped create a clear and strong dissertation, especially because the Influenza modelling results were robust and interpretable. However, this also means that the final framework was not developed as a fully general model for all respiratory viruses. The brief exploratory attempt on RSV showed that the same modelling logic did not transfer equally well without adjustment. This suggests that different viruses may require different modelling assumptions, thresholds, or features.

A further methodological limitation is that the feature engineering process, while meaningful and well grounded, was still restricted to the available surveillance indicators. The study created lag, change, and rolling features from the same basic variables, which proved highly useful. However, future research could consider broader types of engineered features, including longer lag windows, *country*-specific baselines, or additional interaction-based variables.

7.3 Model-Related Limitations

Although the machine learning results were strong, especially for Random Forest, model-related limitations still remain. The first and most obvious point is that **no model achieved perfect prediction**. Even the best model made some incorrect classifications. This means that the framework is useful for outbreak warning, but it does not eliminate uncertainty. In real-world monitoring, there will still be some false alarms and some missed outbreak weeks.

This is important because a model with strong average performance can still make mistakes in critical periods. For example, in an early-warning context, a missed outbreak week may be more important than an error during a low-activity period. Therefore, the performance metrics should be interpreted as useful indicators of overall model quality, but not as proof of perfect reliability in every individual case.

There is also a trade-off between **predictive strength** and **interpretability**. Random Forest was selected as the best-performing model, and it did provide feature importance values, which improved interpretability. However, it is still less transparent than a very simple rule-based system or a basic linear model. This means that while it is interpretable enough for dissertation purposes and useful for analysis, it should still be applied carefully in practical decision-making settings.

7.4 Brief Note on RSV and Pathogen-Specific Differences

A secondary exploratory attempt was made to apply the same general modelling workflow to RSV, but the predictive performance was not as strong as for Influenza. This is an important observation because it suggests that one outbreak framework may not transfer equally well across all respiratory viruses. For this reason, Influenza was retained as the main modelling focus of the dissertation. This decision improved the strength and clarity of the study. RSV is therefore better treated as a note for discussion and future work rather than as a full second modelling result in the main analysis.

7.5 Ethical Considerations

One ethical consideration is that machine learning results can sometimes appear more certain than they really are. A model may achieve high performance metrics, but this should not lead to overconfidence. In a public health setting, predictive models should be used as support tools, not as replacements for professional judgement. The outputs of the model should therefore be understood as helpful signals rather than automatic decisions.

Finally, there is an ethical duty to present the results honestly and transparently. This includes reporting limitations clearly, acknowledging weaker areas such as RSV, and not presenting the framework as more complete than it really is. In that sense, ethical responsibility in this dissertation is closely linked to methodological honesty and careful interpretation.

7.6 Strengths of the Study

One major strength is that the research used an official surveillance dataset, which gives the study direct public health relevance. The framework was not built on artificial or toy data, but on real surveillance indicators that are actually used in monitoring respiratory virus activity.

A second strength is the use of a **chronological train-test split**. This was methodologically stronger than a random split because it better reflected the real forecasting setting. It also made the evaluation design more suitable for a time-based early-warning problem.

A third strength is the comparison of five different machine learning models rather than reliance on a single algorithm. This made the final model selection more credible and allowed the dissertation to demonstrate why Random Forest was chosen as the best model.

CHAPTER 8: CONCLUSION AND FUTURE WORK

This chapter presents the final conclusion of the dissertation and highlights the main contribution of the study. It brings together the findings from the earlier chapters and explains what the research has achieved overall. In addition, it outlines possible directions for future work based on the limitations and observations discussed in the previous chapter.

8.1 Overall Conclusion of the Study

The exploratory analysis showed that the cleaned Influenza sentinel dataset contained clear variation across weeks and countries. Among the core surveillance variables, positivity emerged as the strongest descriptive and predictive signal. The data did not behave like random or static information. Instead, it showed visible periods of lower and higher Influenza activity, making it suitable for outbreak-related analysis.

The machine learning stage confirmed this. All five tuned models performed reasonably well, which demonstrated that the selected surveillance features contained strong predictive information. However, the tuned Random Forest model was identified as the best-performing overall. It achieved the highest cross-validation F1-score, the highest final F1-score, the highest accuracy, and the highest ROC-AUC among the models tested. This means that the final selected model performed strongly not only on the hold-out test set, but also during time-series validation on the training data.

Another major conclusion is that positivity-related behaviour was the most important predictor of next-week outbreak risk. The feature importance analysis of the final tuned Random Forest model showed that current positivity was by far the strongest feature, followed by week and rolling positivity-related context. This suggests that short-term outbreak prediction depends strongly on present virus intensity, recent pattern, and seasonal timing.

Taken together, the findings show that the research aim was successfully achieved. The dissertation developed a tuned and validated machine learning-based framework for next-week's influenza outbreak prediction using official surveillance data, which produced strong, interpretable results.

8.2 Contribution of the Research

From an academic perspective, the study contributes to the area of surveillance-based outbreak analytics by bringing together several important elements in one structured framework. Many studies focus either on descriptive surveillance analysis, general forecasting, or model comparison in isolation. In contrast, this dissertation combines:

- exploratory analysis of surveillance patterns
- feature engineering based on recent trend behaviour
- time-aware validation using time-series cross-validation
- hyperparameter tuning of five machine learning models
- and interpretation of the final selected model

This gives the dissertation a clearer and more integrated contribution than a study that only reports descriptive trends or only compares algorithms without explaining what the data means.

From a practical perspective, the study demonstrates that official surveillance indicators can support more than retrospective reporting. In many settings, surveillance data is used mainly to describe what has already happened. This dissertation shows that the same type of data can also be used for short-term outbreak warning. This has practical value because it suggests that surveillance teams may be able to use recent positivity behaviour and machine learning support to identify increasing outbreak risk earlier.

8.3 Future Work

One possible area for future research is the use of additional external variables. This dissertation focused on the indicators available within the surveillance dataset, which was appropriate for a clear and practical modelling framework. However, future studies could examine whether predictive performance improves when external variables are added, such as:

- weather data,
- mobility indicators,
- vaccination coverage,
- policy-related information,
- or healthcare demand indicators.

A second area for future work is the use of alternative outbreak definitions. In this dissertation, outbreak status was defined using a positivity-based threshold. This was a practical and transparent approach, but future research could test more flexible definitions, such as country-specific baselines, trend-based rules, or multi-indicator outbreak labels.

A third possibility is the extension of the study to different forecast horizons. The present research focused on next-week prediction because it fits the aim of early warning. However, future work could examine two-week, three-week, or longer-range outbreak forecasting to see how far ahead reliable prediction remains possible.

A fourth area for future development is the use of pathogen-specific modelling strategies for other respiratory viruses. The exploratory attempt on RSV suggested that the same framework may not work equally well for all pathogens. This indicates that future studies could build more tailored RSV-specific or multi-pathogen frameworks using different features and threshold logic.

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CHAPTER 10: APPENDIX

Appendix A: Key Data Preparation Code

Import libraries and load the dataset

```
# Import pandas for data handling and analysis
import pandas as pd

# Import numpy for numerical operations
import numpy as np

# Import files module to upload dataset in Google Colab
from google.colab import files

df = pd.read_csv("sentinelTestsDetectionsPositivity.csv")

# Display the first 5 rows of the dataset
print("First 5 rows of the dataset:")
display(df.head())

# Check the number of rows and columns in the dataset
print("Shape of the dataset (rows, columns):")
print(df.shape)

# Display all column names
print("Column names in the dataset:")
print(df.columns.tolist())
```

Filter Influenza sentinel data

```
# Create a copy of the original dataset
df_filtered = df.copy()

# Filter the dataset to keep only primary care sentinel records
df_filtered = df_filtered[df_filtered['survtype'] == 'primary care sentinel']

# Filter the dataset to keep only Influenza records
df_filtered = df_filtered[df_filtered['pathogen'] == 'Influenza']

# Display the first 5 rows of the filtered dataset
print("First 5 rows of the filtered dataset:")
display(df_filtered.head())

# Check the size of the filtered dataset
print("Shape of the filtered dataset (rows, columns):")
print(df_filtered.shape)

# Print a separator line for clean output
print("\n" + "="*60)

# Confirm the surveillance type after filtering
print("Unique surveillance types after filtering:")
print(df_filtered['survtype'].unique())
```

```python
# Confirm the pathogen after filtering
print("Unique pathogens after filtering:")
print(df_filtered['pathogen'].unique())

# Print a separator line
print("\n" + "="*60)

# Check the indicators available after filtering
print("Unique indicators in the filtered dataset:")
print(df_filtered['indicator'].unique())

# Print a separator line
print("\n" + "="*60)

# Check the age categories available after filtering
print("Unique age groups in the filtered dataset:")
print(df_filtered['age'].unique())
```

Reshape indicators

```python
# We use countryname, yearweek, pathogen, and age as the row identifiers
df_pivot = df_filtered.pivot_table(
    index=['countryname', 'yearweek', 'pathogen', 'age'],
    columns='indicator',
    values='value',
    aggfunc='first'
).reset_index()

# Remove the column index name created by pivot_table
df_pivot.columns.name = None

# Display the first 5 rows of the reshaped dataset
print("First 5 rows of the reshaped dataset:")
display(df_pivot.head())

# Check the shape of the reshaped dataset
print("Shape of the reshaped dataset (rows, columns):")
print(df_pivot.shape)

# Print a separator line
print("\n" + "="*60)

# Show the new column names after reshaping
print("Column names after reshaping:")
print(df_pivot.columns.tolist())
```

Create time variables

```python
# Create a copy of the cleaned dataset
df_time = df_clean.copy()

# Extract the year from the 'yearweek' column
df_time['year'] = df_time['yearweek'].str.extract(r'(\d{4})')

# Extract the week number from the 'yearweek' column
df_time['week'] = df_time['yearweek'].str.extract(r'W(\d{2})')

# Convert year and week into integer format
df_time['year'] = df_time['year'].astype(int)
df_time['week'] = df_time['week'].astype(int)

# Create a proper date column using ISO year-week format
df_time['week_start_date'] = pd.to_datetime(
    df_time['year'].astype(str) + '-W' + df_time['week'].astype(str).str.zfill(2) + '-1',
    format='%G-W%V-%u',
    errors='coerce'
)

# Sort the dataset by country and week_start_date
df_time = df_time.sort_values(by=['countryname', 'week_start_date']).reset_index(drop=True)

# Display the first 10 rows to inspect the new time features
print("First 10 rows after creating time features:")
display(df_time[['countryname', 'yearweek', 'year', 'week', 'week_start_date']].head(10))
```

```python
# Check the data types of the new columns
print("Data types of the new time-related columns:")
print(df_time[['year', 'week', 'week_start_date']].dtypes)

# Print a separator line
print("\n" + "="*60)

# Check whether any week_start_date values failed to convert
print("Missing values in week_start_date:")
print(df_time['week_start_date'].isnull().sum())

# Print a separator line
print("\n" + "="*60)

# Check the final shape of the dataset after time feature creation
print("Shape of the dataset after adding time features:")
print(df_time.shape)
```

Create lag features and change based features

```python
# Create a copy of the time-processed dataset
df_features = df_time.copy()

# Create lag-1 feature for positivity
df_features['positivity_lag1'] = df_features.groupby('countryname')['positivity'].shift(1)

# Create lag-1 feature for detections
df_features['detections_lag1'] = df_features.groupby('countryname')['detections'].shift(1)

# Create lag-1 feature for tests
df_features['tests_lag1'] = df_features.groupby('countryname')['tests'].shift(1)

# Create change in positivity from the previous week
# Current positivity minus previous week's positivity
df_features['positivity_change'] = df_features['positivity'] - df_features['positivity_lag1']

# Create change in detections from the previous week
# Current detections minus previous week's detections
df_features['detections_change'] = df_features['detections'] - df_features['detections_lag1']

# Display the first 10 rows with the new features
print("First 10 rows after creating lag and change-based features:")
display(df_features[
    ['countryname', 'yearweek', 'positivity', 'positivity_lag1', 'positivity_change',
     'detections', 'detections_lag1', 'detections_change', 'tests', 'tests_lag1']
].head(10))
```

```python
# Check missing values in the newly created lag features
print("Missing values in lag and change-based features:")
print(df_features[
    ['positivity_lag1', 'detections_lag1', 'tests_lag1', 'positivity_change', 'detections_change']
].isnull().sum())

# Print a separator line
print("\n" + "="*60)

# Check the shape of the dataset after adding lag features
print("Shape of the dataset after adding lag features:")
print(df_features.shape)
```

Create 3-week rolling average features

```
# Create a copy of the feature dataset
df_features2 = df_features.copy()

# Create 3-week rolling average for positivity for each country
df_features2['positivity_roll3'] = (
    df_features2.groupby('countryname')['positivity']
    .transform(lambda x: x.rolling(window=3, min_periods=1).mean())
)

# Create 3-week rolling average for detections for each country
df_features2['detections_roll3'] = (
    df_features2.groupby('countryname')['detections']
    .transform(lambda x: x.rolling(window=3, min_periods=1).mean())
)

# Create 3-week rolling average for tests for each country
df_features2['tests_roll3'] = (
    df_features2.groupby('countryname')['tests']
    .transform(lambda x: x.rolling(window=3, min_periods=1).mean())
)

# Display the first 10 rows with the new rolling features
print("First 10 rows after creating rolling average features:")
display(
    df_features2[
        ['countryname', 'yearweek', 'positivity', 'positivity_roll3',
         'detections', 'detections_roll3', 'tests', 'tests_roll3']
    ].head(10)
)
```

```
# Check missing values in the rolling average features
print("Missing values in rolling average features:")
print(
    df_features2[['positivity_roll3', 'detections_roll3', 'tests_roll3']]
    .isnull()
    .sum()
)

# Print a separator line
print("\n" + "="*60)

# Check the shape of the dataset after adding rolling features
print("Shape of the dataset after adding rolling average features:")
print(df_features2.shape)
```

Create the target variable

```
# Create a copy of the dataset
df_model = df_features2.copy()

# Create the current outbreak flag
# If positivity is 10 or more, mark it as outbreak = 1
# Otherwise, mark it as non-outbreak = 0
df_model['outbreak_current'] = (df_model['positivity'] >= 10).astype(int)

# Create the next-week outbreak target for each country

df_model['outbreak_next_week'] = (
    df_model.groupby('countryname')['outbreak_current'].shift(-1)
)

# Check the first 10 rows to understand how the target was created
print("First 10 rows showing current and next-week outbreak flags:")
display(
    df_model[
        ['countryname', 'yearweek', 'positivity', 'outbreak_current', 'outbreak_next_week']
    ].head(10)
)

# Print a separator line for cleaner output
print("\n" + "="*60)

# Check missing values in the target column
# The last week of each country will usually have missing next-week target
print("Missing values in 'outbreak_next_week':")
print(df_model['outbreak_next_week'].isnull().sum())
```

```
# Remove rows where next-week outbreak target is missing
df_model = df_model.dropna(subset=['outbreak_next_week']).reset_index(drop=True)

# Convert the target column to integer format
df_model['outbreak_next_week'] = df_model['outbreak_next_week'].astype(int)

# Check class distribution of the final target variable
print("Class distribution of next-week outbreak target:")
print(df_model['outbreak_next_week'].value_counts())

# Print a separator line
print("\n" + "="*60)

# Check the shape of the final modelling dataset
print("Shape of the modelling dataset after target creation:")
print(df_model.shape)
```

Create a chronological train-test split

```
# Create a copy of the final modelling dataset
df_split = df_final.copy()

# Sort the dataset by the weekly start date
# This ensures the split follows the actual time order
df_split = df_split.sort_values(by='week_start_date').reset_index(drop=True)

# Define the feature columns again for modelling
feature_columns = [
    'tests',
    'detections',
    'positivity',
    'year',
    'week',
    'positivity_lag1',
    'detections_lag1',
    'tests_lag1',
    'positivity_change',
    'detections_change',
    'positivity_roll3',
    'detections_roll3',
    'tests_roll3'
]

# Define the target column
target_column = 'outbreak_next_week'
```

```
split_index = int(len(df_split) * 0.8)

# Create training feature set
X_train = df_split[feature_columns].iloc[:split_index]

# Create testing feature set
X_test = df_split[feature_columns].iloc[split_index:]

# Create training target set
y_train = df_split[target_column].iloc[:split_index]

# Create testing target set
y_test = df_split[target_column].iloc[split_index:]

# Display the shapes of the train and test sets
print("Training feature set shape:")
print(X_train.shape)

print("\n" + "="*60)

print("Testing feature set shape:")
print(X_test.shape)

print("\n" + "="*60)

print("Training target set shape:")
print(y_train.shape)
```

Appendix B: Machine Learning Code

Time-series cross-validation and tuned Logistic Regression

```
import pandas as pd

from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import GridSearchCV, RandomizedSearchCV, TimeSeriesSplit
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from sklearn.metrics import roc_auc_score, confusion_matrix, classification_report

# Reset index to make row-based CV splits clean and safe
train_data = train_data.sort_values(by='week_start_date').reset_index(drop=True)
test_data = test_data.sort_values(by='week_start_date').reset_index(drop=True)

# Recreate X and y
X_train = train_data[feature_columns].copy()
X_test = test_data[feature_columns].copy()
y_train = train_data[target_column].copy()
y_test = test_data[target_column].copy()

# Create time-series cross-validation splits using full weeks
unique_train_weeks = np.array(sorted(train_data['week_start_date'].unique()))
tscv = TimeSeriesSplit(n_splits=5)

cv_splits = []

for train_week_idx, val_week_idx in tscv.split(unique_train_weeks):
    train_weeks_cv = unique_train_weeks[train_week_idx]
    val_weeks_cv = unique_train_weeks[val_week_idx]

    train_rows = train_data.index[train_data['week_start_date'].isin(train_weeks_cv)].to_numpy()
    val_rows = train_data.index[train_data['week_start_date'].isin(val_weeks_cv)].to_numpy()

    cv_splits.append((train_rows, val_rows))
```

```
# Create a helper function to evaluate each tuned model on the test set
def evaluate_tuned_model(model_name, fitted_model, X_test, y_test):
    # Predict class labels
    y_pred = fitted_model.predict(X_test)

    # Predict probabilities for ROC-AUC
    y_prob = fitted_model.predict_proba(X_test)[:, 1]

    # Calculate evaluation metrics
    accuracy = accuracy_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred, zero_division=0)
    recall = recall_score(y_test, y_pred, zero_division=0)
    f1 = f1_score(y_test, y_pred, zero_division=0)
    roc_auc = roc_auc_score(y_test, y_prob)

    # Print the performance
    print(f"{model_name} Performance:")
    print(f"Accuracy  : {accuracy:.4f}")
    print(f"Precision : {precision:.4f}")
    print(f"Recall    : {recall:.4f}")
    print(f"F1-Score  : {f1:.4f}")
    print(f"ROC-AUC   : {roc_auc:.4f}")
```

```python
# Apply Grid Search with time-series cross-validation
logreg_search = GridSearchCV(
    estimator=logreg_pipeline,
    param_grid=logreg_param_grid,
    scoring=scoring_metric,
    cv=cv_splits,
    n_jobs=-1,
    verbose=1
)

# Train the tuned Logistic Regression model
logreg_search.fit(X_train, y_train)

# Get the best tuned model
logreg_pipeline = logreg_search.best_estimator_

print("Best Logistic Regression Parameters:")
print(logreg_search.best_params_)

print("\n" + "="*60)

print("Best Cross-Validation F1-Score:")
print(round(logreg_search.best_score_, 4))

print("\n" + "="*60)

# Evaluate on the test set
y_pred_logreg, y_prob_logreg, logreg_accuracy, logreg_precision, logreg_recall, logreg_f1, logreg_roc_auc = evaluate_tuned_model(
    "Logistic Regression",
    logreg_pipeline,
    X_test,
    y_test
)

# Store the best CV score
logreg_cv_f1 = logreg_search.best_score_
```

Tuned Decision Tree

```python
# Create the Decision Tree model
dt_model = DecisionTreeClassifier(random_state=42)

# Define the hyperparameter search space
dt_param_dist = {
    'criterion': ['gini', 'entropy'],
    'max_depth': [3, 5, 10, 15, None],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4],
    'class_weight': [None, 'balanced']
}

# Apply Randomized Search with time-series cross-validation
dt_search = RandomizedSearchCV(
    estimator=dt_model,
    param_distributions=dt_param_dist,
    n_iter=20,
    scoring=scoring_metric,
    cv=cv_splits,
    n_jobs=-1,
    verbose=1,
    random_state=42
)

# Train the tuned Decision Tree model
dt_search.fit(X_train, y_train)

# Get the best tuned model
dt_model = dt_search.best_estimator_

print("Best Decision Tree Parameters:")
print(dt_search.best_params_)
```

```python
print("Best Cross-Validation F1-Score:")
print(round(dt_search.best_score_, 4))

print("\n" + "="*60)

# Evaluate on the test set
y_pred_dt, y_prob_dt, dt_accuracy, dt_precision, dt_recall, dt_f1, dt_roc_auc = evaluate_tuned_model(
    "Decision Tree",
    dt_model,
    X_test,
    y_test
)

# Store the best CV score
dt_cv_f1 = dt_search.best_score_
```

Tuned Random Forest

```python
# Create the Random Forest model
rf_model = RandomForestClassifier(
    random_state=42,
    n_jobs=-1
)

# Define the hyperparameter search space
rf_param_dist = {
    'n_estimators': [100, 200, 300, 400],
    'max_depth': [5, 10, 15, 20, None],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4],
    'max_features': ['sqrt', 'log2', None],
    'class_weight': [None, 'balanced']
}

# Apply Randomized Search with time-series cross-validation
rf_search = RandomizedSearchCV(
    estimator=rf_model,
    param_distributions=rf_param_dist,
    n_iter=20,
    scoring=scoring_metric,
    cv=cv_splits,
    n_jobs=-1,
    verbose=1,
    random_state=42
)

# Train the tuned Random Forest model
rf_search.fit(X_train, y_train)

# Get the best tuned model
rf_model = rf_search.best_estimator_

print("Best Random Forest Parameters:")
print(rf_search.best_params_)
```

Tuned Support Vector Machine (SVM)

```
# Create a pipeline for Support Vector Machine
svm_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('model', SVC(probability=True, random_state=42))
])

# Define the hyperparameter search space
svm_param_dist = {
    'model__C': [0.1, 1, 10, 50],
    'model__kernel': ['linear', 'rbf'],
    'model__gamma': ['scale', 0.01, 0.1, 1],
    'model__class_weight': [None, 'balanced']
}

# Apply Randomized Search with time-series cross-validation
svm_search = RandomizedSearchCV(
    estimator=svm_pipeline,
    param_distributions=svm_param_dist,
    n_iter=12,
    scoring=scoring_metric,
    cv=cv_splits,
    n_jobs=-1,
    verbose=1,
    random_state=42
)

# Train the tuned SVM model
svm_search.fit(X_train, y_train)

# Get the best tuned model
svm_pipeline = svm_search.best_estimator_

print("Best SVM Parameters:")
print(svm_search.best_params_)
```

```
print("Best Cross-Validation F1-Score:")
print(round(svm_search.best_score_, 4))

print("\n" + "="*60)

# Evaluate on the test set
y_pred_svm, y_prob_svm, svm_accuracy, svm_precision, svm_recall, svm_f1, svm_roc_auc = evaluate_tuned_model(
    "Support Vector Machine",
    svm_pipeline,
    X_test,
    y_test
)

# Store the best CV score
svm_cv_f1 = svm_search.best_score_
```

Tuned k-nearest neighbours (KNN)

```
# Create a pipeline for K-Nearest Neighbours
knn_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('model', KNeighborsClassifier())
])

# Define the hyperparameter grid
knn_param_grid = {
    'model__n_neighbors': [3, 5, 7, 9, 11, 15],
    'model__weights': ['uniform', 'distance'],
    'model__p': [1, 2]
}

# Apply Grid Search with time-series cross-validation
knn_search = GridSearchCV(
    estimator=knn_pipeline,
    param_grid=knn_param_grid,
    scoring=scoring_metric,
    cv=cv_splits,
    n_jobs=-1,
    verbose=1
)

# Train the tuned KNN model
knn_search.fit(X_train, y_train)

# Get the best tuned model
knn_pipeline = knn_search.best_estimator_

print("Best KNN Parameters:")
print(knn_search.best_params_)
```

```
print("Best Cross-Validation F1-Score:")
print(round(knn_search.best_score_, 4))

print("\n" + "="*60)

# Evaluate on the test set
y_pred_knn, y_prob_knn, knn_accuracy, knn_precision, knn_recall, knn_f1, knn_roc_auc = evaluate_tuned_model(
    "K-Nearest Neighbours",
    knn_pipeline,
    X_test,
    y_test
)

# Store the best CV score
knn_cv_f1 = knn_search.best_score_
```

ROC curves

```
# Import the required tools for ROC curve plotting
from sklearn.metrics import roc_curve, auc
import matplotlib.pyplot as plt

# Calculate ROC curve points for Logistic Regression
fpr_logreg, tpr_logreg, _ = roc_curve(y_test, y_prob_logreg)
roc_auc_logreg = auc(fpr_logreg, tpr_logreg)

# Calculate ROC curve points for Decision Tree
fpr_dt, tpr_dt, _ = roc_curve(y_test, y_prob_dt)
roc_auc_dt = auc(fpr_dt, tpr_dt)

# Calculate ROC curve points for Random Forest
fpr_rf, tpr_rf, _ = roc_curve(y_test, y_prob_rf)
roc_auc_rf = auc(fpr_rf, tpr_rf)

# Calculate ROC curve points for Support Vector Machine
fpr_svm, tpr_svm, _ = roc_curve(y_test, y_prob_svm)
roc_auc_svm = auc(fpr_svm, tpr_svm)

# Calculate ROC curve points for K-Nearest Neighbours
fpr_knn, tpr_knn, _ = roc_curve(y_test, y_prob_knn)
roc_auc_knn = auc(fpr_knn, tpr_knn)

# Create the ROC curve figure
plt.figure(figsize=(10, 7))

# Plot ROC curves for all tuned models
plt.plot(fpr_logreg, tpr_logreg, label=f'Logistic Regression (AUC = {roc_auc_logreg:.4f})')
plt.plot(fpr_dt, tpr_dt, label=f'Decision Tree (AUC = {roc_auc_dt:.4f})')
plt.plot(fpr_rf, tpr_rf, label=f'Random Forest (AUC = {roc_auc_rf:.4f})')
plt.plot(fpr_svm, tpr_svm, label=f'SVM (AUC = {roc_auc_svm:.4f})')
plt.plot(fpr_knn, tpr_knn, label=f'KNN (AUC = {roc_auc_knn:.4f})')
```

```
# Plot the diagonal reference line
plt.plot([0, 1], [0, 1], linestyle='--')

# Add title and labels
plt.title('ROC Curve Comparison of All Tuned Machine Learning Models')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')

# Add legend
plt.legend(loc='lower right')

# Adjust layout
plt.tight_layout()

# Show the plot
plt.show()
```